

MAXIMISING EXAM MARKS IN **Senior maths**

INTERMEDIATE Book 1

*Top tips for
avoiding
common errors!*



BARBARA LYNCH



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Introduction

Every year, examiners, teachers and assessors in every state produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. *Maximising Exam Marks in Senior Maths* is a collection of reproducible worksheets which are designed to assist senior students in those areas that they have traditionally found difficult.



Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

THINGS TO KNOW – Content that students should learn and be able to apply

HINTS – Helping students approach questions correctly

TRICKS – Giving short-cuts or easier ways to attack questions

SLIPS – Providing opportunities for students to practise working carefully to avoid common errors

TRAPS – Providing the types of contexts in which students make common errors

The table of contents below shows the topic areas used in each activity. Pages which stress appropriate examination and revision techniques are also included. The worksheets could be integrated into the learning process or used as part of examination or test revision. In most cases, an electronic calculator would be useful and where appropriate, a graphical calculator can be used.

The CD-ROM attached to the inside back cover contains a PDF version of this book.

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1 – Tricks and traps in arithmetic

Most mathematics problems will involve arithmetic in one form or another. Scientific calculators and graphical calculators are only as good as the person pressing the buttons! Sometimes it is better not to use a calculator.



THINGS TO KNOW

- ☞ Decimal approximations are not accepted if an exact answer is required.
- ☞ Estimating the answer can help you to check that the calculator answer is sensible.
- ☞ Answers should be given to the correct degree of accuracy (decimal places or significant figures).
- ☞ Rounding numerical answers too early can give the wrong answer. Calculations within the question should be stored as you go and not rounded. Only the final number should be rounded if so directed.
- ☞ In practical problems there are usually units of measurement, so include the correct units in the answer.

- ① Find the exact value of $59\,774\,957\,682 \times 3$.

HINT

Too long for your calculator display? Multiply by hand?

- ② Find the exact value.

A $\frac{3}{4} \times \frac{1}{8} \times 10\frac{2}{3}$

HINT

Treat as fractions and remember $10\frac{2}{3} = \frac{32}{3}$.

B $10 - \frac{2}{3} \div \frac{2}{9}$

HINT

To divide by a fraction, 'invert and multiply'. Careful of order of operations.

- ③ Find an *exact* answer for $8^{-1} + 12^{-1}$.

- ④ When I used a calculator to find $\sqrt{256}$, I got 15.

- A Is this answer about the right size? _____
 B For $\sqrt{256}$, what should I expect the last digit to be?

C The inverse of $\sqrt{\quad}$ is square. Is $15^2 = 256$?

D Was my first answer correct? _____

- ⑤ Find the value of the following, giving your answers correct to 2 significant figures.

A $2\pi \times \sqrt{\frac{52}{9.8}}$ _____

B $\frac{-15 + \sqrt{15^2 - 4 \times 2 \times 7}}{4}$

THINGS TO KNOW

Rules for counting significant figures

- ☞ Count all zeros within a number as significant. (2305 and 2.005 have 4 sig figs)
- ☞ Count all zeros to the right of the decimal point as significant. (56.00 and 5.050 have 4 sig figs)
- ☞ Do not count zeros that keep the decimal point in place as significant. (3200 has only 2 sig figs)

Rules for rounding

- ☞ All digits after the rounding digit are dropped.
- ☞ If the digit to the right of the rounding digit is equal to or more than 5, add 1 to the rounding digit.



- ⑥ Evaluate $\sqrt{\frac{367.2}{6.4 \times 2.8}}$. What is the difference between the answers you will give if you are asked for 'correct to 2 significant figures' and 'correct to 2 decimal places'?

- ⑦ Use first digit rounding to find whether $5.94 \times 6\,371 - 35\,253$ should give a positive or negative answer. Use a calculator to check your answer, explain the result and suggest an alternative method of checking.

TRAP

First digit rounding of large numbers can give wrong answers!

- ⑧ Use a calculator to find

A $\sqrt{3} + 2$ and store the answer in memory.

B $2.6 \times \pi$.

C Hence find $\frac{2.6 \times \pi}{\sqrt{3} + 2}$ correct to 2 decimal places.

TRICK

Notice that you worked out the denominator first and had to put it in memory so that the fraction would be the right way up.
Round at the end!

- ⑨ Evaluate this functions question:
 $3 \times [2 \times (3)^2 - 3] - 2 \times [3 \times 3 - 4]^2 + [3 \times 3 - 4]$.

THINGS TO KNOW

Order of operations check!

Brackets are first,
Of is second,
Division and Multiplication,
then Addition and Subtraction,
working left to right.

- ⑩ A trigonometry question gave:
 $2.4 \times [(4.153)^2 - 2] - 3 \times [2.7^3 - 3]^2 + [7.1^3 + 3]$.
Give your answer correct to 4 significant figures.

- ⑪ Evaluate $(7 \times 10^5)^2 \div (4 \times 10^{-6})^{\frac{1}{2}} \times (2.4 \times 10^4)$.
Give your answer in scientific notation.

THINGS TO KNOW

Scientific notation consists of a number greater than 1 and less than 10, multiplied by a power of 10.

- ⑫ Evaluate $\frac{2}{7} \times (\frac{4}{9} \times (\frac{3}{2})^2) + (\frac{6}{5} \div \frac{7}{10})$. Give an exact answer.

2 – Tricks and traps in algebra

Get to know your standard Algebra rules, such as the quadratic formula for solving equations, factorising and expansion rules, and absolute values. Remember that squaring equations can cause traps with signs. When you are finishing an easy question, you are more likely to make a silly algebraic mistake than when you are finishing a harder one, so it is best to slow down and concentrate.



- ① I think the simplification of $\frac{5}{x+2} - \frac{4x}{3}$ is $\frac{(4x-5)(x-3)}{3(x+2)}$.

To test this is correct, I will substitute a number for x in each, and see if both give the same answer.

- Ⓐ Which is the easiest number to substitute out of 0, ± 1 , ± 2 and ± 3 ? _____
- Ⓑ Is my simplification correct? _____
- Ⓒ Which number out of 0, ± 1 , ± 2 and ± 3 should I **not** substitute? Why?

- ② For $x^2 + y^2 = 9$, make y the subject of the equation.

- ③ You are solving the following equation and have just finished factorising.

$$\sqrt{x+6} = x$$

$$x+6 = x^2$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

- Ⓐ Find the values of x for which the last line of the equation is true.

- Ⓑ Substitute these x values into the original equation.
Are they both solutions? Why?

- ④ Copy and complete this factorisation: $x^3 - 64 = (x-4)(x^2 \text{ _____ } 16)$

- ⑤ Copy and complete this sum to give a whole number answer.

$$\sum_{n=2}^5 (3n+1) = (3 \times 2 + 1) + (3 \times 3 + 1) + \text{_____} + \text{_____} = \text{_____}$$

- ⑥ Ⓐ Copy and complete this expansion: $(x+y)^2 = x^2 \text{ _____}$

- Ⓑ Expand and simplify: $(a+b)^2 - 4ab$.

- Ⓒ Hence complete $(a+b)^2 - 4ab = \text{_____}$
 $= (\text{_____})^2$.

TRAP

When you square equations,
substitute back to check
there are no extra solutions!

THINGS TO KNOW

Know the rules!



- 7 Expand $(4x - 16)^2$ by taking out a common factor first.

$$\begin{aligned}(4x - 16)^2 &= [4(x - 4)]^2 \\ &= \underline{\hspace{2cm}} (x - 4)^2 \\ &= \underline{\hspace{2cm}} (\underline{\hspace{2cm}}) \\ &= \underline{\hspace{2cm}}\end{aligned}$$

TRICK

4 is squared too!

- 8 Graphs are helpful in solving some inequations. The graph represents the equation $y = (x + 2)(x - 1)(4 - x)$ and it cuts the x -axis at the points where $x = -2, 1, 4$.

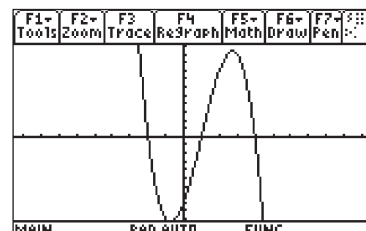
- A Use this information to find the values of x for which $(x + 2)(x - 1)(4 - x) > 0$.

- B Use the graph to show that $y \leq 0$.

- 9 It is necessary to simplify expressions in many calculus questions.

- A Simplify $\frac{(x + h)^3 + (x + h) - (x^3 + x)}{h}$.

- B Hence find $\lim_{h \rightarrow 0} \frac{(x + h)^3 + (x + h) - (x^3 + x)}{h}$.



- 10 Absolute value or 'mod' can cause problems and provide many opportunities to slip up. The best way to avoid these slips is to choose easy methods.

- A Find values of x for which $|x - 2| \leq 1$ by writing it in the form $-1 \leq x - 2 \leq 1$.

- B Find values of x for which $|x - 2| \leq 1$ by writing it in the form $+(x - 2) \leq 1$ or $-(x - 2) \leq 1$.

_____ or _____

- C Find values of x for which $|x - 2| \leq 1$ by writing it in the form of an equation: $|x - 2| = 1$.

$$+(x - 2) = 1 \text{ or } -(x - 2) = 1$$

_____ or _____

$$x = \underline{\hspace{2cm}} \text{ or } x = \underline{\hspace{2cm}}$$

Use graphs of $y = |x - 2|$ and $y = 1$ to see where $|x - 2| \leq 1$.

So _____ or _____ are the solutions.

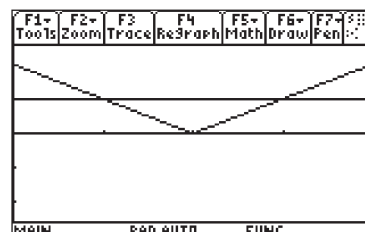
- D Which is the easiest way for you to solve this inequation?

SLIP

+2 to both sides.

TRAP

+2 to both sides, and be careful of the minus changing the less than sign to greater than.



3 – Expanding and factorising – best way

Factorising is writing an expression as a product of its factors. It is one way of simplifying an expression and is a necessary skill in the solution of many mathematical problems. Factorising over Q gives factors containing only rational numbers, while factorising over R gives factors containing irrational numbers as well. Expanding usually means multiplying out brackets. If you know about the Binomial Theorem, you can use that instead of multiplying. Some calculators can be used to factorise and expand expressions. The main thing is to be careful of slips in your work and practise, practise, practise.



- ① Expand $(3a - 1)^3$. _____
- ② Find the term in x of the expansion $(x - 3y^2)^4 = x^4 + 4 \times x^3 \times (-3y^2) + 6 \times x^2 \times (-3y^2)^2 +$ _____.

- ③ Divide $4y^3 + 2y^2 + 6y - 1$ by $2y + 3$.

HINT

What method will you use? Long division?
Synthetic division? Fractions?

THINGS TO KNOW

Common factor first – always!

Two terms:

- ☞ Difference of perfect squares: $a^2 - b^2 = (a - b)(a + b)$
- ☞ Difference of perfect cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- ☞ Sum of perfect cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Three terms:

- ☞ Perfect square: $a^2 + 2ab + b^2 = (a + b)^2$
 $a^2 - 2ab + b^2 = (a - b)^2$
- ☞ Quadratic trinomial using cross method or splitting the middle term for the distributive law in reverse
- ☞ Quadratic trinomial using completing the square
- ☞ Quadratic trinomial using substitution to make it a simpler quadratic trinomial
- ☞ Quadratic trinomial using completing the square with the middle term and then using the difference of two squares
- ☞ Quadratic trinomial using the factor theorem
- ☞ Quadratic trinomial using the quadratic formula
The general quadratic expression $ax^2 + bx + c$ has factors given by:

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a} \right) \left(x + \frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a} \right)$$

More than three terms:

- ☞ group terms two and two – so that there is a factor common to each pair
- ☞ group terms three and one – where the three terms form a perfect square and so there is the difference of perfect squares
- ☞ factor theorem (the factors of the constant help find the factors)



3 – Expanding and factorising – best way (cont'd)

- 4 Factorise, where possible, each of the following over the real field, R .

A $2x + 4y - 18z$ _____

B $16a^2 - 4b^2$ _____

C $z^2 - 5$ _____

D $3x^3 + 24y^3$ _____

E $p^2 + q^2$ _____

- 5 Factorise each of the following over the real field, R .

A $9y^2 - 12y + 4$ _____

B $2x^2 + x - 6$ _____

C $x^2 - 6x + 4$ _____

D $x^4 - 5x^2 + 6$ _____

E $x^3 + 2x + 12$ _____

- 6 Can the expression $3x^2 - 5x + 2$ be factorised over the real field? Give reasons for your answer. If it factorises, what sort of factors does it have?

- 7 Factorise over the real field, R .

A $x^3 - 8$ _____

B $2x^2 - 5x - 4$ _____

C $x^2 + 6xy + 9y^2 - 4z^2$ _____

D $x^3 + x^2 - 22x - 40$ _____

- 8 Write the expression $2y^3 + 11y^2 + 2y - 15$ in the form $(y - 1)(ay^2 + by + c)$ and hence find linear factors over the rational field.

- 9 Find linear factors of $ax^3 - bx$ over the real field, where $a, b \in R^+$.

TRICK

Common factors should be taken out before any other method is applied.

HINT

You can check your factors are right by multiplying them out.

HINT

If you are using the factor theorem, the factors of the constant term help you find the factors to test for.

THINGS TO KNOW

These don't have numbers in them and you may find that more difficult. Think of the a and b as if they were numbers and look at the key words:

- 'linear' (from line graph) means that there are no squares, cubes, etc. of x in the answer;
- real field and a and b are real positive numbers means that you can have roots in the answer.

4 – Indices and logs

You need to know your index and log laws if you are going to be able to simplify expressions without making silly mistakes.



THINGS TO KNOW

Definition: If $x = a^n$ for $a \in \mathbb{R}^+ \setminus \{1\}$ then $\log_a x = n$

Index laws

$a, b \in \mathbb{R} \setminus \{0\}$ and $m, n \in \mathbb{R} \setminus \{0\}$

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$\frac{1}{a^n} = a^{-n}$$

$$(ab)^n = a^n b^n$$

$$a^0 = 1$$

$a^{\frac{1}{q}} = {}^q\sqrt{a}$ or the positive q th root of a
and/or the positive square root of a

$a^{\frac{p}{q}} = {}^q\sqrt{a^p}$ or the q th root of a to the power p

Log laws

Since for $a \in \mathbb{R}^+ \setminus \{1\}$ the logarithm of zero or of a negative number is not defined.

If $0 < a < 1$, the logarithms of a to any base will be negative.

$$\log_a(x \times y) = \log_a(x) + \log_a(y)$$

$$\log_a(x \div y) = \log_a(x) - \log_a(y)$$

$$\log_a(x)^y = y \log_a(x)$$

$$\log_a\left(\frac{1}{x}\right) = -\log_a x$$

$$\log_a 1 = 0$$

$$\text{Change of base: } \log_a x = \frac{\log_b x}{\log_b a}$$

- ① Complete the following sentence:

$e \approx 3$ and $3^2 = 9$ so $e^2 \approx$ _____ and $\log_e 9 \approx$ _____.

HINT

Be careful with algebra.

- ② Simplify: $x(0.2)^{-1} = \frac{x}{0.2} =$ _____

- ③ Use a calculator to find $\log 15 - \ln 15$ correct to 3 decimal places.

THINGS TO KNOW

Log is log base 10 but ln is log base e .

- ④ Find the exact value of $e^{\log_e 9}$.

HINT

Definition!

- ⑤ (A) Find the exact value of k for which $4^x = e^{kx}$ for all $x \in \mathbb{R}$.

TRAP

- ☞ x should not be in the answer to the first part.
- ☞ 'Hence' is an important word; it means use your answer to Part (a).
- ☞ Show all your working and remember that $(\log 3) \times x \neq \log(3 \times x)$.

- (B) Hence simplify $\log_e [(4)e^{kx}]$.

- ⑥ If $y = 10 \log_{10} \left(\frac{x}{10^{-12}} \right)$, find y when $x = 100$.

HINT

Index and log laws!



- 7 The approximate value of $e = 2.71828$. You solved the equation $e^x = 17$ and you got the answer $x = 2.5$. Is that a reasonable answer?

- 8 If $a^n = b^{n+m}$, show that $n = \frac{\log b^m}{\log(\frac{a}{b})}$.

HINT

Log both sides, use the log laws and be careful with your algebra!

- 9 Use the change of base rule $\log_3 6 = \frac{\log_e 6}{\log_e 3}$ to find $\log_3 6$ correct to

A 2 decimal places.

B 2 significant figures.

- 10 Find the **exact** value of k for which $2^x = e^{kx}$ for all $x \in \mathbb{R} \setminus \{0\}$.

TRICK

Take log base e both sides.

- 11 Factorise.

A $(e^x)^2 + e^{3x}$

B $e^{2x} - 6e^x + 8$

HINT

Treat $e^{2x} = (e^x)^2$ as if it were an x^2 in a quadratic.

- 12 Is $(e^x)^2 = e^{x^2}$, $(e^x)^2 > e^{x^2}$ or is $(e^x)^2 < e^{x^2}$ when x is positive?

HINT

Graphs can be helpful, or substitute numbers.

- 13 Solve for x : $4e^{kx} = 3$

- 14 Solve for x : $(\frac{1}{e})^x > \frac{1}{2}$

THINGS TO KNOW

Log both sides but remember that when dividing by a negative in an inequation, you have to swap the inequality sign.

- 15 Use a graph or otherwise to find the values of x for which

A $\log_e x^2 \leq 0$.

B $2\log_e x \leq 0$.

C $\log_e x \leq x$.

THINGS TO KNOW

Remember for $\log_a x$, x can't be zero or negative.

5 – Solving equations and inequations

The solution of equations is a necessary skill in Mathematics. It is particularly useful for finding intercepts when drawing graphs. Many algebraic equations can be solved by factorising and then using the 'null factor law': if the product of any two numbers is zero, then one or both of the numbers is zero. Absolute value functions and inequations require care with the positive and negative signs. The main thing is to be careful of slips in your work. Remember that another word for 'solution' of an equation is 'root' of the equation.



- ① Solve for p : $-8 = 4p$.

SLIP

Answer is not $-\frac{1}{2}$.

- ② Solve for x .

A $-1 \leq \frac{x}{4} \leq 1$

B $-1 \leq \frac{x}{-4} \leq 1$

- ③ Solve for x : $\frac{3}{x-2} < 1$.

- ④ Solve for x : $3x^2 + 2x - 8 = 0$.

- ⑤ Find exact values of x such that $x^3 + 2x^2 - 4x - 8 = 0$.

- ⑥ The discriminant for a quadratic equation is $4 - 4 \times \frac{1}{3} \times -15$. What does this mean for the nature and number of solutions to this equation?

- ⑦ Find the real value(s) of y for which $y^4 - 4y^2 + 4 = 0$.

TRAP

Multiply everything and watch the signs.

TRAP

There are plenty of traps in this question:

- ☞ Errors in Algebra; you can't square inequalities without thought.
- ☞ Under what conditions can you multiply both sides by $x - 1$?
- ☞ Not using a graph
- ☞ Only expecting one solution

HINT

Use the easiest method to factorise: factor theorem or grouping or trial and error?

THINGS TO KNOW

For the equation $ax^2 + bx + c = 0$

- ☞ real solutions exist if $b^2 - 4ac \geq 0$
- ☞ there is one solution if $b^2 - 4ac = 0$
- ☞ there are two real solutions if $b^2 - 4ac > 0$
- ☞ there are two rational solutions if $b^2 - 4ac$ is a perfect square
- ☞ no real solutions exist if $b^2 - 4ac < 0$
- ☞ the discriminant $\Delta = b^2 - 4ac$.

TRICK

How many terms?
What form of equation is it?



- 8 You are asked to use Algebra to solve the simultaneous equations below.

$$2y = 3x + 4$$

$$2x^2 + 3x - 4y = -6$$

so you substituted for y to get:

$$2x^2 + 3x - 2(3x + 4) = -6$$

Find x and y .

- 9 If $y = x^3 + 5x^2 + 8x + 4$, write y in the form:

$$y = (x + 1)(\text{_____}).$$

Hence find the exact solutions of the equation $x^3 + 5x^2 + 8x + 4 = 0$.

HINT

The main thing is to get the right quadratic factor. You can use long division or synthetic division or perhaps work it out in your head.

- 10 Solve for a .

A $|a| = 4$

B $|x + 2| = |x|$

THINGS TO KNOW

Use $\begin{cases} |x| = x \text{ for } x \geq 0 \\ |x| = -x \text{ for } x < 0 \end{cases}$ or $|x| = \sqrt{x^2}$
and be careful of your signs.

- 11 Solve the inequation for y .
 $y^2 + 3y < 0$.

HINT

Perhaps a graph will help.

- 12 Find the value of x for which
 $2x - 5 \leq x + 7 \leq 4x + 1$.

TRICK

Split into two separate inequations, the left one and the right one.

- 13 Find the value of x for which $\frac{x+a}{b} = c$, where a , b and c are positive constants.

6 – Exponential and logarithmic functions

The exponential and log functions deserve special mention because of the restrictions on their domains and ranges, and the relationship between them.

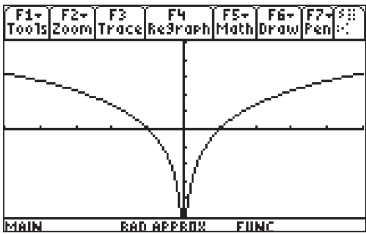
THINGS TO KNOW

$f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = e^x$ and
 $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ where $f(x) = \log_e x$.



1 For a log function $f(x) = \log_a N$, what are the restrictions on a and N ?

2 This is a graph of $y = \log_e x^2$ from a graphical calculator.



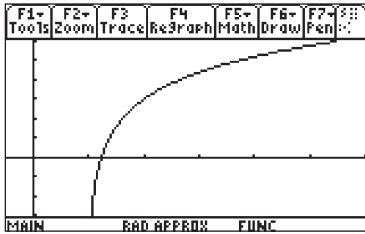
HINT

Know your graphs!
They will always help you.

A What is wrong with it? How would you draw it on the exam paper?

B State the x and y intercepts where they exist.

3 The graph for the function with rule $f(x) = 2\log_e(x - 1) + 3$ is shown.



A State the domain and range of the function.

B Find the inverse function.

C State the domain and range of the inverse function.

4 The rule for the inverse function of $f(x) = 6 - 3e^{-x}$ can be found by making y the subject of the equation $e^{-y} = \frac{6-x}{3}$. Show that the rule for the inverse is $f^{-1}(x) = \log_e\left(\frac{3}{6-x}\right)$. Show all your working.

HINT

You will need to take logs of both sides of the exponential equation and then use the log laws. But be careful of the negative signs and take short simple steps in your algebra so you don't make silly mistakes.

5 For the function $f: (2, \infty) \rightarrow \mathbb{R}$ where $f(x) = 0.2\log_e(x - 2)$

A state the domain and range.

B give reasons why the inverse function f^{-1} exists.

C find the rule for the inverse function $f^{-1}(x)$.

TRAP

There are plenty of traps in this question. Know your interval notation if it's in the syllabus (you should check):

(a, b) is $a < x < b$

$[a, b]$ is $a \leq x \leq b$

$[a, b)$ is $a \leq x < b$

$[a, b]$ is $a \leq x \leq b$

Make sure you use the domain that is given in the question. Be careful of your algebra and make sure that your explanations are clear.

6 If $P(x) = e^{2x - 4x^2}$, for what value of x does P start to get smaller?

HINT

Remember, e^x will start to get smaller when x goes through its maximum.

7 – Growth and decay

For growth and decay problems, you need to be able to interpret word problems and be accurate with your index and logarithm algebra. Be careful!



- ① A model of the decay of a material is given by $Y = Ae^{-kt} + Y_0$ where $A = 100$.

- Ⓐ Find the values of Y_0 and k , correct to 2 decimal places, by filling in and then solving the simultaneous equations as follows:

When $t = 0$, $Y = 70$ so _____ (1)

When $t = 20$, $Y = 15$ so _____ (2)

- Ⓑ What value would you expect for Y when $t = 10$, correct to 2 decimal places?

Use the unrounded values of Y_0 and k to get the most accurate answer. _____

- ② $P = 4e^{0.02t}$ where $t \geq 0$.

- Ⓐ Find the rule for t in terms of P .

- Ⓑ Find the value of t correct to 2 significant figures for which P will be double its initial value.

- ③ If $P = 4e^{(0.02t - 0.0001t^2)}$, the rate of change of P is given by: $R = 0.08(1 - 0.001t)e^{(0.02t - 0.0001t^2)}$. Find the rate of change of P when $t = 10$. Give your answer correct to 3 decimal places.

- ④ A model of population growth is given by the equation $Y = Y_0 e^{f(t)}$ where Y is in thousands of millions and t is the number of years after 1980. Give decimal answers correct to 3 decimal places.

- Ⓐ What will be the value of t in 1990? _____

- Ⓑ If the population when $f(t)$ is zero is 100 million, what value will you substitute for Y_0 ? _____

- ⑤ A model for fuel consumption, L tonne per 1000 kilometres, for a ship increases exponentially, but at low speeds there is an idling factor added.

$$L = 4e^{0.004s} + \frac{10}{s+3}, \text{ where } s \text{ km/h is the speed.}$$

- Ⓐ What is the restriction of the values of s ? _____

- Ⓑ When the ship is stationary, what is the value of the idling factor? _____

- Ⓒ If you found L to be 4.93 when s was 10, how many tonnes of fuel would the ship use in 500 km?

- ⑥ A model for the amount of current, I microamps, that continues to flow t seconds after a switch is turned off is

HINT

Make sure that you round carefully.

TRAP

Remember, you are making t the subject of the formula, **not** finding the inverse function.

HINT

Understanding what is required is helped by underlining key words when you read the question. Take note of words like 'initial' and 'stationary', what the variables are, and check the units. Try to have a mental picture of the situation.



given by the equation $I = -k(t^2 - 10t + 28)e^{\frac{t}{2}} + 12$, where k is a constant.

(A) How would you find the initial current? _____

(B) What would you need to know if you wanted to find the value of k ?

(C) If you wanted to find the current flowing after half a minute, what value of t would you substitute? _____

7 Two different bacteria, A and B , can infect chicken pieces. To kill these bacteria you need to cook the chicken at a temperature of at least 120°C . The number of living bacteria in a piece of chicken after t minutes at this temperature can be modelled using the equations:

$$A(t) = 1200e^{-0.6t} \text{ for } A\text{-type bacteria}$$

$$B(t) = 1000e^{-0.4t} \text{ for } B\text{-type bacteria.}$$

Give all decimal answers correct to 3 decimal places.

(A) How long will it take to kill half the A -type bacteria?

(B) What is the total number of live bacteria (types A and B) when the cooking started, $t = 0$?

(C) How many bacteria will be left after the chicken has been cooked for 5 minutes?

(D) What percentage of the total number of bacteria have been killed by the cooking in the first 5 minutes?

(E) At what time will there be equal numbers of the two types of bacteria in the chicken?

8 (A) The number of sales of mobile phones every three-month period is to be modelled using the equation $N = ab^t$ where $t \geq 0$ and a and b are positive constants. Is this likely to be a good model? Why?

(B) If the graph of a function that models the percentage of people owning a laptop computer t years after their introduction has an inflection point at $(3.75, 27.3)$, what does this mean?

(C) If the graph of a function that models the number of rabbits that can be kept in an enclosure shows a gradual increase until it asymptotes to the value 8, explain the effect.

HINT

For a model to represent a real situation accurately, it has to give sensible answers for all values of the variable. Remember that an exponential increases at an increasing rate.

8 – Functions of functions of functions of ...

You will need to know and understand the definitions of functions and their inverses, composites (functions of functions), and domains and ranges. What is most important is that you do not spoil a good answer with sloppy algebra.



- ① In finding the inverse function of the function with rule $f(x) = 2x - 7$, the student interchanged x and y to get $x = 2y - 7$.
Complete the following: $f^{-1}(x) =$ _____
- ② If the graph of a 1:1 function has an asymptote with equation $x = 2$, what is the equation of the asymptote of the graph of $y = f^{-1}(x)$, the inverse function?
- _____

- ③ A Find the inverse of the function with rule $f(x) = x^2$.

B Why is the inverse not a function?

C How could you restrict the domain of f so that the inverse would be a function?

SLIP

Remember the positive and negative signs when you take the square root. Also be careful of brackets, common denominators and substitutions.

D Sketch the graph of $y = x^{\frac{1}{2}}$.

TRICK

$y = x^{\frac{1}{2}}$ is a tricky graph!

- ④ A function has the rule $f(x) = \frac{3}{x+2} - \frac{2x}{7}$ so

$$f(x-1) = \frac{3}{(x-1)+2} - \frac{2(x-1)}{7}.$$

Find $f(x-1)$ in simplest form.

TRAP

Positive and negative signs, brackets, common denominators, careful substitutions.

- ⑤ For $f(x) = x^3 - 2x^2$, find each of the following in simplest form.

A $\frac{1}{f(x)}$ _____

B $f(\frac{1}{x})$ _____

C $f(x^{-1})$ _____

D $[f(x)]^{-1}$ _____

E An expression that could be used to find $f^{-1}(x)$ _____

- ⑥ If $f(x) = 3x$ and $g(x) = x^2$

A find $f[g(x)]$. _____

B find $g[f(x)]$. _____

C does it matter in which order you find the composite function? _____

- ⑦ If $f(x) = 2x^2 - x$ and $g(x) = 3x - 4$

A find $3f(x)$. _____

B find $f[g(x)]$ in simplest form. _____



Ⓒ Hence find in simplest form $3f(x) - f[g(x)]$. _____

Ⓓ Find in simplest form $3f(2g(x))$. _____

8 Explain why the function with rule $y = \sqrt{8x}$ is a function. _____

9 Explain why the function $f: R \setminus \{0\} \rightarrow R$ where $f(x) = \frac{1}{x^2}$ has no inverse function. _____

10 In finding the inverse function of the function with rule $f(x) = (x + 2)^2$, the student interchanged x and y to get $x = (y + 2)^2$. Complete the following:

A The inverse of $f(x)$ is / is not a function because _____

B An inverse function of $f, f^{-1}(x)$, can be formed by restricting the domain to values of x : _____ or _____

C The rule for an inverse function of $f, f^{-1}(x)$, formed by restricting the domain would be _____ for x _____ or _____ for x _____.

11 Identify the following statements as true or false for a function f and its inverse function f^{-1} .

A Domain of f = Range of f^{-1} (True / False)

B Range of f = Domain of f^{-1} (True / False)

C Domain of f = Domain of f^{-1} (True / False)

D Range of f = Range of f^{-1} (True / False)

E Every function has an inverse function. (True / False)

F The graphs of f and f^{-1} are reflections of each other in the line $y = x$. (True / False)

12 For the function with rule $f(x) = x^2 + 4x + 2$, the point of intersection of the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$ can be found by solving the equations of $f(x)$ and $f^{-1}(x)$ simultaneously **or** by solving the equations of $f(x)$ and $y = x$ simultaneously. Use the method you find easier to find this point of intersection for $x < -1$.

13 In a practical problem, two relationships

$$y_1 = 30 + 25 \sin\left(\frac{(4t-1)\pi}{2}\right); 0 \leq t \leq 2\pi$$

$$y_2 = -75t + 40; t \leq 24$$

are to be sketched and then added to give the graph of $y_1 + y_2$. What will be the domain of the graph of $y_1 + y_2$? How did you find it?

HINT

Practical problems can have 'yucky' expressions but the question isn't necessarily difficult.

9 – Transformation of graphs

The order in which transformations are carried out makes a difference. You should always remember that when $y = f(x)$ is transformed to $y = \pm a.f[n(x - b)] + c$, you need to know the basic shape and use the order of transformation dilation (a and n), reflection ($\pm$ in front) and then translation (b and c), working from left to right.



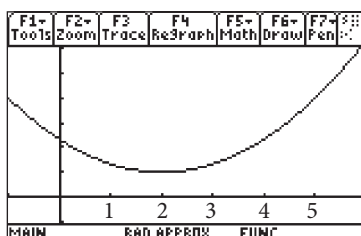
- ① The graph with equation $y = 3x^2$ touches the x -axis at the origin. Write the equation of the graph which has exactly the same shape as $y = 3x^2$ but

(A) touches the x -axis at the point $(2, 0)$. _____

(B) has a turning point at the point $(0, 2)$. _____

- ② The graph with equation $y = m(x - n)^2 + 1$ is shown.

The turning point is at the point $(2, 1)$ and the y -intercept is at $(0, 2\frac{1}{3})$. Find the values of n from the graph and m by substitution.



HINT

You can't read off the value of m like you can the value of n . You need to calculate it.

- ③ What sequence of transformations maps the graph with equation $y = f(x)$ onto the graph with equation $y = f(\frac{x}{3}) + 1$?

HINT

What does the dilation $\frac{1}{3}x$ mean here? You need to make sure that you understand this fully so that you can give a clear answer.

- ④ Read off the transformations you would need to get the graph with equation

(A) $y = \frac{-2}{x+1} - 2$ from the graph with equation $y = \frac{1}{x}$.

(B) $y = \log_e(x + 2) - 5$ from the graph with equation $y = \log_e x$.

(C) $y = 2 \sin 2(x - \pi)$ from the graph with equation $y = \sin x$.

(D) $y = -3e^{2x-4}$ from the graph with equation $y = e^x$.

- ⑤ (A) The graph with equation $y = x^3$ is translated -4 units parallel to the x -axis and then dilated by a factor of $\frac{1}{2}$ from the y -axis. What is the equation of the transformed graph?

TRAP

Careful: is it $(2x____)$ or is it $2(x____)$?

(B) The graph with equation $y = x^3$ is dilated by a factor of $\frac{1}{2}$ from the y -axis and then translated -4 units parallel to the x -axis. What is the equation of the transformed graph?

- ⑥ If $f(x) = \cos x$ is transformed to $g(x) = 2\cos(3x - \pi) + 1$, what transformation is needed?



9 – Transformations of graphs (cont'd)

- 7 What are the equations of the asymptotes of the graphs with the following equations?

- A $y = \frac{2}{x-1} + 4$ _____
 B $y = 3e^{2x} + 1$ _____
 C $y = 2\log_e(3x + 2) - 6$ _____

HINT

A sketch graph for the transformed shape should give you the asymptotes. Remember, you want their equations.

- 8 Using the graph of the function $f(x) = x^3$, the graph of $g(x)$ is obtained with a translation of +2 in the x -direction. Using the graph of $g(x)$, the graph of $h(x)$ is obtained with a translation of -1 in the y -direction. Using the graph of $h(x)$, the graph of $j(x)$ is obtained with a dilation with scale factor $\frac{1}{3}$ from the y -axis. Write the rules for $g(x)$, $h(x)$ and $j(x)$.

HINT

You need to use the previous result, even though the word 'hence' is not used. Consequential errors would still give you full marks.

- 9 The graph of $f(x) = (x - 2)^2(x + 1) - 1$ has turning points at (0, 3) and (2, -1). What are the turning points of the graph with equation $y = f(\frac{x}{3}) - 1$?

TRAP

Remember that $f(\frac{x}{3}) \neq \frac{f(x)}{3}$. Notice also that the original graph has the translation -1 in it too.

- 10 You are asked to sketch the following graphs. Simplify each expression and hence describe the resulting graph.

- A $y = e^3 e^x$ _____

 B $y = \log_e(x) + \log_e(3)$ _____

HINT

Simplifying can make graphing possible.

- 11 A function f is defined by the rule $f(x) = 3 - 2e^{-x}$ for $x \geq 0$. Sketch the graph of f , clearly labelling any points of intersection with the axes, and any asymptotes with their equations.

HINT

Using a graphical calculator is good for learning, but of no use here! You need to understand the concepts.

- 12 For the curve whose equation is $y = -2x - 1 + \frac{6}{(x+1)^2}$, write the equations of the asymptotes and indicate on a sketch how the curve approaches these asymptotes. No other information is required on the graph.

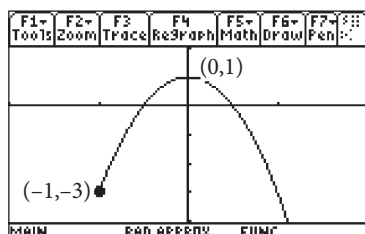
10 – Intervals and restricted graphs

You may need to understand interval notation. Alternatively, the domains can be restricted using inequality notation. Either way it is important that graphs are drawn according to the domain given.

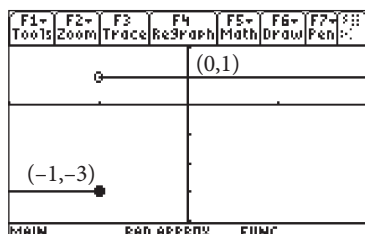


- ① Write the domain of the function for each graph.

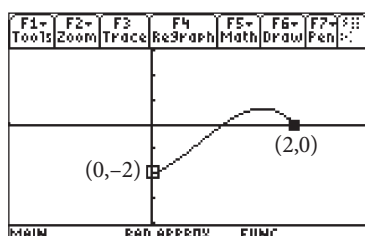
A _____



B _____

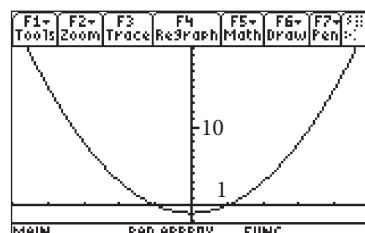


C _____



- ② The graph of the function $f: R \rightarrow R$ where $f(x) = x^2 - 1$ is shown. Redraw the graph of f if the domain is

- A $f: [2, \infty) \rightarrow R$.
 B $f: (2, \infty) \rightarrow R$.
 C $f: [2, 4] \rightarrow R$.
 D $f: (2, 4) \rightarrow R$.



- ③ Sketch the graph of the function $f(x) = 6 - 3e^{-x}$ for $x \geq 0$.

THINGS TO KNOW

Note the domain and range, label the graph well and show the asymptote clearly. Watch out for the shape of the graph if you are using a graphical calculator.

- ④ For a function to be differentiable throughout its domain, its graph must be smooth and continuous.

- give the equation
- state if the graph is smooth
- state if the graph is continuous.

Hence state which functions are differentiable throughout the domain.

THINGS TO KNOW

'Differentiable' means able to be differentiated and so there is a tangent line.

'Smooth' means has no sharp points or kinks.

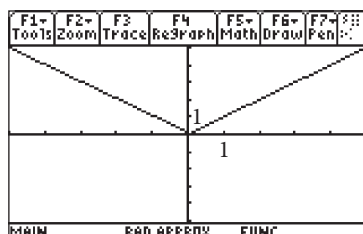
'Continuous' means has a value for all values of x with no holes or jumps.

Of course you need $\lim_{x \rightarrow c} f(x) = f(c)$ for all c as well, but you don't need to worry about that.

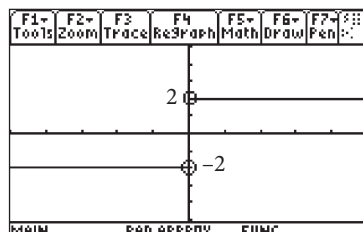


10 – Intervals and restricted graphs (cont'd)

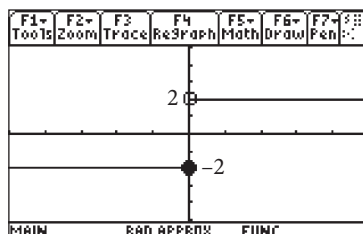
A _____



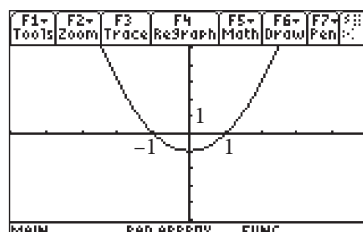
B _____



C _____



D _____



5 Sketch the graph of

$$y = \begin{cases} x^2 - 1 & x \leq 2 - 1 \\ \frac{1}{x} + 1 & -1 < x < 0 \\ \cos x & x > 0 \end{cases}$$

and hence write the domain and range of f .

TRAP

Remember to draw on one set of axes and then just read off the domain and range. But watch for traps caused by values being excluded like $x, y \neq 0$. Also remember that $\cos$ must be in radians.

6 Find the domain and range of the function with the rule below.

$$y = \begin{cases} \frac{-2}{x} & -\infty < x < 0 \\ 2x^2 & 0 < x < 2 \\ e^{-0.5x} & 2 \leq x \leq \infty \end{cases}$$

HINT

Make sure you draw the graph so you can see what's going on.

7 Sketch the graph of the function
 $f: (-2\pi, 2\pi) \rightarrow \mathbb{R}, f(x) = \tan\left(\frac{x}{4}\right) + 1$.
 Note that $(-2\pi, 2\pi)$ means $-2\pi < x < 2\pi$.

THINGS TO KNOW

Remember to

- make sure you have answered in the right domain
- check that you have the right curve shape (approaching asymptote, etc.).

Check that you have

- marked the x-intercept correctly (use π , not 3.14)
- written in the equations of the asymptotes.

The graphics calculator will make the graph look like the asymptotes are joined together, so be careful!

11 – Trigonometric functions and graphs

The calculator can be your friend and your enemy when you are dealing with trig functions.



THINGS TO KNOW

When you are graphing trig functions with a calculator, remember that:

- ☞ you can find the value of the function – as long as you are in the right mode (radians or degrees)
- ☞ you can copy the graph from the calculator screen – as long as you get the correct intercepts, make it 'curvy' not 'wedgy' and make your asymptotes look right
- ☞ it's sometimes handy to use $\pi \approx 3$ to check your answers.

- ① Explain how any angle between 0° and 360° can be converted into radians.

- ② Convert the angles in degrees to angles in radians, in terms of π .

A 225°

B 330°

- ③ Use a calculator to evaluate, correct to 3 decimal places.

A $\sin 15^\circ - \sin \frac{\pi}{2}$

B $\cos 75^\circ + \tan 1$

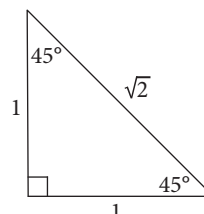
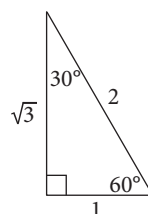
- ④ If $x = 2$, find the value to 4 significant figures.

A $\sin(x) \cdot x$

B $\sin x^2$

- ⑤ You should know your special angles and their sin, cos and tan. Practise by writing them into this table.

Special Angles	0° 0c	30° $\frac{\pi^c}{6}$	45° $\frac{\pi^c}{4}$	60° $\frac{\pi^c}{3}$	90° $\frac{\pi^c}{2}$
Sine					
Cosine					
Tangent					



THINGS TO KNOW

The word 'explain' means that you need to give a definition and to give reasons so that it is easily understood.

TRAP

$\sin 15^\circ$ and $\cos 75^\circ$ are in degrees but $\sin \frac{\pi}{2}$ and $\tan 1$ are in radians.



11 – Trigonometric functions and graphs (cont'd)

6 If $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, what are the values of the following?

A $\sin \frac{2\pi}{3}$ _____

B $\sin \frac{4\pi}{3}$ _____

C $\sin \frac{-\pi}{3}$ _____

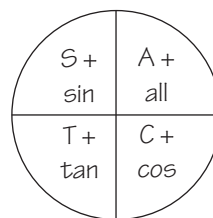
7 If $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$, find $\sec \frac{\pi}{3} + \cot \frac{\pi}{6}$.

8 Solve the equation for α : $\sin \alpha = \frac{1}{\sqrt{2}}$; $0 \leq \alpha \leq 2\pi$.

9 Find the sum of the solutions for x .

$$\sin\left(\frac{x}{2}\right) = \frac{1}{2}; 0 \leq x \leq 4\pi$$

10 Sketch the graph of one period of $y = -4 \sin 2x$.



THINGS TO KNOW

One 'co' between each pair:

$$\sec x = \frac{1}{\cos x} \quad \operatorname{cosec} x = \frac{1}{\sin x} \quad \cot x = \frac{1}{\tan x}$$

HINT

Read the question – you want the sum of the solutions. Make sure you are on the right domain of 0 to 4π .

HINT

Label the axes, only draw one period and use radian measure on the calculator.

11 The depth of water t hours after low tide is given by d metres where $d(t) = 4 - 2\sin\left(\frac{2\pi t}{23} + \frac{\pi}{2}\right)$. If low tide was at 2 a.m., use a graph or otherwise to find

A the value of t for the next low tide. _____

B the time of the next low tide. _____

HINT

Remember, $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$ or otherwise could be the period of d .

12 – Polynomial functions and their graphs

Know the facts about the properties of polynomial graphs! The domain is $\mathbb{R}$. The graph will generally have one less turning point than the degree of the polynomial ($y = x^3$ has 2, $y = x^4$ has 3, etc.). Odd numbers of repeated factors may give inflection points. Repeated roots give 'touchers'.



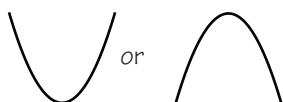
THINGS TO KNOW

Degree 1: linear function (straight line)

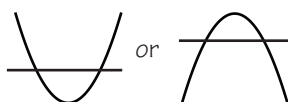
$y = mx + c$ has gradient m and y -intercept c

Degree 2: quadratic function (parabola)

$y = \pm a(x - b)^2 + c$ has turning point at (b, c) and y -intercept at $(0, c \pm ab^2)$

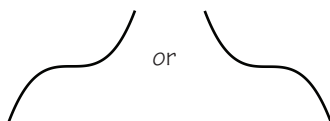


$y = \pm a(x - h)(x - k)$ has x -intercepts at $x = h$ and $x = k$

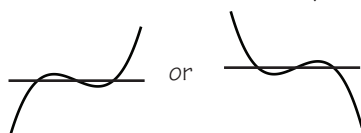


Degree 3: cubic function

$y = \pm a(x - b)^3 + c$ has inflection point at (b, c)

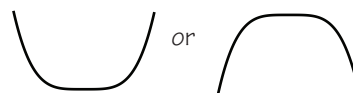


$y = \pm a(x - h)(x - k)(x - \ell)$ has x -intercepts at $x = h$, $x = k$ and $x = \ell$

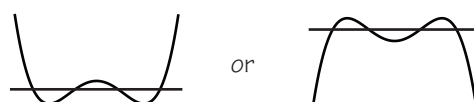


Degree 4: quartic function

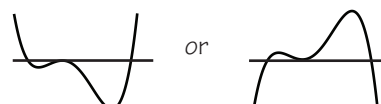
$y = \pm a(x - b)^4 + c$ has turning point at (b, c) .



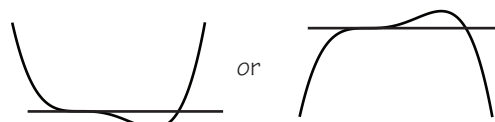
$y = \pm a(x - h)(x - k)(x - \ell)(x - m)$ has x -intercepts at $x = h$, $x = k$, $x = \ell$ and $x = m$



$y = \pm a(x - h)^2(x - k)(x - \ell)$ has x -intercepts at $x = h$ (touch), $x = k$ and $x = \ell$



$y = \pm a(x - h)^3(x - k)$ has x -intercepts at $x = h$ (inflection) and $x = k$



- ① If the graph of the function $f(x) = ax + c$ has intercepts on the axes at $(3, 0)$ and $(0, 2)$, find the values of a and b and hence the gradient of the graph.

- ② Show that the graph with equation $y = 2x^2 - 4x + 3$ and the graph with equation $y = 2(x - 1)^2 + 1$ are identical. State their turning point and y -intercept.

HINT

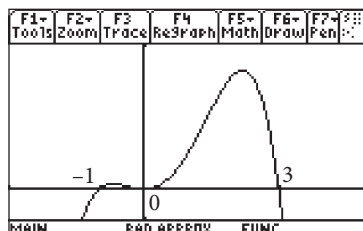
'Show' means put in all the steps.
Try multiplying out, it's easiest!
Watch for algebraic errors though.
Get turning point from one form
and intercept from the other.



12 – Polynomial functions and their graphs (cont'd)

- ③ The graph of a cubic function $y = f(x)$ has turning points at $x = 3$ and $x = \frac{1}{3}$.
What will be the turning points of the graph of the function $y = f(x + 2)$?

- ④ Write a possible equation for this graph.



- ⑤ The graph of the function $y = f(x)$ touches the x -axis at the point $(b, 0)$ and cuts the x -axis at the point $(c, 0)$.
If $f(x)$ is a polynomial of degree 3, find a possible equation of the graph.

- ⑥ For the function $f(x) = ax^3 + bx^2 + cx + d$, the graph passes through the points $(0, 10)$, $(5, 0)$, $(1, -12)$ and $(-2, 0)$.

- A Write this information as four linear equations.

- B Hence solve to find $f(x)$.

HINT

Be careful that electronic answers are all correct – you might get zero, for example.

- ⑦ For the function $f(x)$ you are given the following clues.

- $f(0) = 0$, $f(1) = 2$ and $f(3) = 0$
- the gradient of the graph is zero at $x = 0$ and $x = 2$
- the gradient is negative for $x > 2$ and positive or zero everywhere else

- A Is the function a cubic or a quartic? _____

- B Draw a possible graph of f .

HINT

Make sure you use all the information.

HINT

Be careful of your algebra and make sure you use the information from part a.

Know the properties of turning points in cubics and quartics.

- C If $f(x) = px^3(x - q)$, find the values of p and q .

13 – Other graphing practice

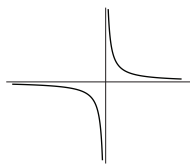
Make sure you have lots of graphing practice before you attempt this mixed set of questions. Remember that reciprocal is 'one over ...', so the reciprocal of x is $\frac{1}{x}$ and such graphs are never continuous; the function $f(x) = \frac{1}{x}$ doesn't exist when the denominator is zero. Adding ordinates is a useful skill that also uses reciprocal graphs. Finally, there is more practice sketching trigonometric and polynomial graphs.



THINGS TO KNOW

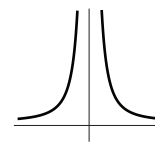
$$y = \frac{1}{x} \text{ or } y = x^{-1}$$

Rectangular hyperbola with vertical asymptote $x = 0$ and horizontal asymptote $y = 0$



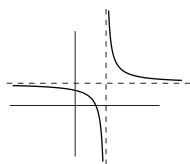
$$y = \frac{1}{x^2} \text{ or } y = x^{-2}$$

Truncus with vertical asymptote $x = 0$ and horizontal asymptote $y = 0$



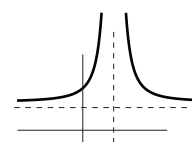
$$y = \frac{a}{x-h} + k$$

Rectangular hyperbola with vertical asymptote $x = h$ and horizontal asymptote $y = k$



$$y = \frac{a}{(x-h)^2} + k$$

Truncus with vertical asymptote $x = h$ and horizontal asymptote $y = k$



- ① Sketch the graph of

A $y = \frac{3}{x-2} + 2$.

B $y = \frac{1}{(x+1)^2} - 2$.

- ② If $y_1 = x^2$ and $y_2 = \frac{1}{(x-1)^2}$, sketch the graph of each on the same axes and hence use addition of ordinates or otherwise to sketch the graph of $y = x^2 + \frac{1}{(x-1)^2}$.
What are the equations of the asymptotes?

- ③ A Determine the equations of the asymptotes of the curve $y = \frac{x^3 - 5x + 1}{x^2 - 9}$.

- B State in words (or indicate clearly on a sketch) the behaviour of the curve as it approaches each asymptote.

- C Find the coordinates of any points of intersection of the curve with an asymptote. There is no need to sketch the whole graph.

- ④ Sketch the graph of one period of $y = \tan x + 1$.

HINT

Make sure you draw only one period, clearly show the asymptotic behaviour (the graphical calculator curves the graph towards the asymptotes), put in the exact values of the intercepts with + and - signs, and make sure your asymptotes are labelled with their equations.

HINT

Remember that when you are sketching graphs, any asymptotes should be drawn and labelled and the coordinates of some points marked to demonstrate the shape.

TRAP

Asymptotes and discontinuous curves can cause problems, particularly when you use a calculator.



- 5 Sketch the graph of $y = 2\sin[3(t - \frac{\pi}{6})]$; $-\pi \leq t \leq \pi$.
Hence find how many solutions there are to the equation $y = 2\sin[3(t - \frac{\pi}{6})] = 1$ in the domain.

TRAP

There are many traps here!

Make sure you are in the right domain, in radian mode not degrees, that you show the correct phase, amplitude and period. Curvy not 'wedgy'! Note 'hence' and answer the question asked – don't solve the equation.

- 6 A cosine graph has an amplitude of 2 and period $\frac{1}{5}$. Complete the following:

$$f(t) = \underline{\hspace{2cm}} \cos(\underline{\hspace{1cm}} t).$$

HINT

Know your transformations!

- 7 For the function with rule $f(x) = (x - 2)^2(x + 1) + 1 = k$, find the range of values of k for which this equation has more than one solution.

TRICK

Draw the graph of the function $f(x) = (x - 2)^2(x + 1)$ and then move it up one to get the equation given.

Now draw a horizontal line representing the value k and see what happens as you move it up and down.

- 8 For the function with rule $f(x) = (x - 2)^2(x + 1)$

- A find the values of x for which $f(x) > 0$

- B find $f(x + h)$

- C find the values of h for which there is only one positive solution to the equation $f(x + h) = 0$.

HINT

Watch your setting out and make sure your argument is clear. Looking at the graphs will help. Of course $f(x + h)$ will result in the graph moved h units to the left. Only one positive solution means roots will have opposite signs – you might like to use that.

- 9 A Sketch the graph of $y = f(x)$ where $f(x) = \frac{10}{1 + 2e^{-x}}$ for $-5 \leq x \leq 5$.

HINT

Make sure that

- ☞ the scale of your graph is clear
- ☞ the curves are accurate and well-drawn
- ☞ the equations of asymptotes are written on the graph
- ☞ the coordinates of the inflection points are marked clearly.

- B Where are the asymptotes?

- C Find the coordinates of the point of inflection and mark them clearly on the graph.

14 – Tricks and traps in differentiation

Differentiation is a skill like adding and subtracting. You need to know how to do it for many calculus problems. Different examinations provide different formula sheets, but you should still learn them by heart.



- ① Many of these standard differentiations will be on your list to learn. Cross out the ones you don't need and write down in your own notation the ones you need to learn.

HINT

You may be given a formula sheet but you should still know the rules so you recognise them when you see them.

THINGS TO KNOW

Standard derivatives:

$$\frac{d}{dx}(x^n) = nx^{n-1}; n \in \mathbb{R} \setminus \{0\}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\log_e x) = \frac{1}{x}; \frac{d}{dx}(\log_e kx) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin ax) = a \cos ax$$

$$\frac{d}{dx}(\cos ax) = -a \sin ax$$

$$\frac{d}{dx}(\tan ax) = a \sec^2 ax$$

Product rule:

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx};$$

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

Quotient rule:

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2};$$

$$\frac{d}{dx}\left\{\frac{f(x)}{g(x)}\right\} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx};$$

$$\frac{d}{dx}f(g(x)) = f'(g(x)) = f'(g(x))g'(x)$$

Approximation:

$$f(x+h) \approx f(x) + hf'(x)$$

Note: $\ln x = \log_e x$; $x > 0$

- ② Find the derivative of:

A $x^2 - 3x^3 + 17x - 5$

B e^{15x}

C $\sin 5x$

- ③ State the rule that has been used in each of the following.

A $\frac{d}{dx}[3x^4 \log_e(x+2)] = [\log_e(x+2)] \frac{d}{dx}(3x^4) + (3x^4) \frac{d}{dx}[\log_e(x+2)]$ where $x > -2$

B $\frac{d}{dx}[e^{x^2}] = 2x \cdot e^{x^2}$

C $\frac{d}{dx}\left(\frac{x^2}{x-1}\right) = \frac{(x-1) \cdot 2x - x^2 \cdot 1}{(x-1)^2}$ where $x \neq 1$

TRAP

Careful! Answer the question!



- ④ Finding the derivative using rules such as product, quotient and chain rule (function of a function) leaves lots of room for slip-ups. **Do not simplify** after you have differentiated these.

HINT

Be careful you choose the right rule and watch the brackets! If you use a calculator, the examiner will know! Be careful of $\ln(3x)$.

A $(3x^4 - 4)^3$

B $(\frac{1}{x} + 2x)^4$ where $x \neq 0$

C $\frac{x^2 + 2}{\ln x + 1}$ where $x > 0$

D $(\sqrt{x} + 2)(\ln(3x) + 2x)$ where $x > 0$

E $\frac{5e^{0.02t}}{6.5e^{0.3t-15}}$ where $t \geq 0$

- ⑤ Differentiate the following.

A $x^2 \cos 5x$ _____

B $5m(1 - m)^8$ _____

C $2e^{3x^2+x}$ _____

D $\log_e(3t - 1)$ _____

E $\frac{2}{3x^2 + 1}$ _____

- ⑥ Is the derivative of the function $f(x) = e^{2x}$ equal to:

A $2e^{2x} f'(e^{2x})$ $2e^{2x} f'(x)$ $2f'(e^{2x})$ or $f'(e^{2x})$

B $2e^{2x} f(e^{2x})$ $2f(x)$ $2f(e^{2x})$ or $f(e^{2x})$

- ⑦ For $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = (x - 2)^2(x + 1) + 1$,
 $f'(x) = (x - 2)(ax + b)$ where a and b are constants.
 Use **Calculus** to find a and b .

SLIP

Notice they are not all with respect to x .
 Does $(1 - m)^6 = (m - 1)^6$?

SLIP

Hint: Differentiate the given $f(x)$, write it in the same form as the given $f'(x)$ and then equate to given a and b . Use calculus but you can check with your calculator.
 Also, you need to decide whether you will multiply it out or use the product rule.

15 – Gradients and differentiation

Differentiation of a function gives an expression for the gradient of the graph at any point. The gradient of the secant or chord of a graph can be found from first principles using the formula for the gradient of a line joining two points.



THINGS TO KNOW

The gradient of a line joining points (x_1, y_1) and (x_2, y_2) is given by: $m = \frac{y_2 - y_1}{x_2 - x_1}$

- ① The gradient of the line joining points $(x, 2x^2 - 1)$ and $[(x + 1), 2(x + 1)^2 - 1]$ is $\frac{[2(x + 1)^2 - 1] - [2x^2 - 1]}{(x + 1) - x}$. Show that the gradient can be written as $4x + 2$.

SLIP

Careful of brackets!

- ② If $f(x) = 4 - 2x^2$ then complete the following to find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[4 - 2(x+h)^2] - (4 - 2x^2)}{h}$$

SLIP

Careful of brackets!

$$= \frac{4 - 2(x^2 + 2xh + h^2) - 4 + 2x^2}{h}$$

$$= \frac{4 - 2x^2 - 4xh - 2h^2 - 4 + 2x^2}{h}$$

$$= \frac{-4xh - 2h^2}{h}$$

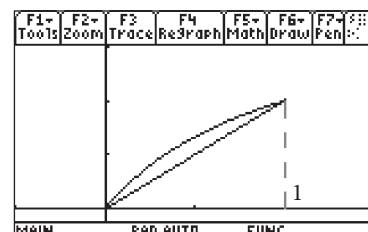
- ③ If $f(x) = \frac{2x}{x-3}$, find $f'(2)$ from first principles.

SLIP

It's easier to substitute the 2 at the beginning. Be careful that

- ☞ $2x$ becomes $2(x+h)$ or $2(2+h)$, not $2x+h$ or $4+h$
- ☞ the algebraic manipulation of the fractions is correct
- ☞ you put the limit sign in the right place (on the RHS of the equals sign)
- ☞ you take the limit carefully.

- ④ The graph shows $y = f(x)$ where $f(x) = \frac{2x}{x+1}$ for $0 \leq x \leq 1$; that is, the interval $[0, 1]$. The gradient of the chord shown is 1.



- A What is the average rate of change of $f(x)$ in the interval $[0, 1]$?

- B Draw a second chord with gradient of approximately $f'(1)$.

- C Use calculus to find

$$f'(x).$$

$$f'(1).$$

HINT

What does a gradient of approximately $f'(1)$ mean? 'Use calculus' does not mean use a calculator, but of course you can use a calculator to check an answer.

- ⑤ A The approximate value of $\sqrt{9.2}$ can be found using Calculus with $f(x+h) \approx f(x) + hf'(x)$.

So if $f(x) = \sqrt{x} = x^{\frac{1}{2}}$, use $x = 9$ and complete the following.

THINGS TO KNOW

Approximation:

$$f(x+h) \approx f(x) + hf'(x)$$



$$f(x) = \sqrt{x} = x^{\frac{1}{2}}$$

$$f'(x) = \underline{\hspace{2cm}}$$

$$f(9) = \underline{\hspace{2cm}}$$

$$f'(9) = \underline{\hspace{2cm}}$$

$$f(9+h) = f(9) + hf'(9)$$

$$f(9.2) = f(9) + 0.2f'(9)$$

$$= \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

B State the approximation of $\sqrt{9.2}$ correct to **I** 2 decimal places **II** 4 significant figures.

$\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$

6 **A** If $f(x) = x^5$, state

$$f(x+h) \cdot \underline{\hspace{2cm}}$$

$$f(x+0.1) \cdot \underline{\hspace{2cm}}$$

$$f(x-0.1) \cdot \underline{\hspace{2cm}}$$

B The formula $f(x+h) \approx f(x) + hf'(x)$ is used to find an approximate value of a function where h is small.

If $f(x) = x^5$, find an approximation for

I $(3.1)^5 \cdot \underline{\hspace{2cm}}$

II $(2.9)^5 \cdot \underline{\hspace{2cm}}$

8 For the curve with equation $y = \frac{2x^2+1}{x^2+3x+2}$ ($x \neq -1, 2$), the gradient is given by $\frac{dy}{dx} = \frac{(x^2+3x+2) \cdot 4x - (2x^2+1)(2x+3)}{(x^2+3x+2)^2}$.

Simplify this derivative and hence find the exact values of x for which the gradient of the curve is zero.

$\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$

9 **A** Find the exact value of k for which $2^x = e^{kx}$.

$\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$

B Hence use calculus to find the derivative of $y = 2^x$.

$\underline{\hspace{2cm}}$
 $\underline{\hspace{2cm}}$

10 Each of the following have at least one solution to the equation $f'(x) = -0.5$ within the given domain. How many solutions can be expected?

A $f(x) = (x-1)^2 - 2$ where $0 \leq x \leq 3$ $\underline{\hspace{2cm}}$

B $f(x) = (x-1)^3 - 2x$ where $0 \leq x \leq 3$ $\underline{\hspace{2cm}}$

C $f(x) = e^{2x} - 3x$ where $0 \leq x \leq 1$ $\underline{\hspace{2cm}}$

D $f(x) = \sin(x) + \cos(2x)$ where $0 \leq x \leq 2\pi$ $\underline{\hspace{2cm}}$

TRAP

Careful of negatives! Check answers with a calculator.

HINT

Be careful of the brackets and notice the words 'hence' and 'exact'.

TRAP

$\log_e$ both sides but be careful of your mathematical notation and your algebra. Remember $\log_e(k)x$ is different from $\log_e(kx)$. 'Hence use calculus' means use what you just found, and you must use calculus. Of course you can check with your calculator, but show all your working. What rule will you use to differentiate?

TRAP

It's $f'(x)$ not $f(x)$!
 What does the graph of $f'(x)$ look like?

16 – Applications of differentiation to tangents to curves

Not only do you need to know how to differentiate various functions, you also need to understand how the derivative relates to the gradient of the tangent to the curve of the function, and to be able to find the equation of this tangent at a given point. The gradient of the tangent to $y = f(x)$ at the point (x_1, y_1) is given by $f'(x_1)$ and the gradient of the normal by $\frac{-1}{f'(x_1)}$.



- ① Find $f'(2)$ if $f(x) = 2x^3 - x$.

- ② For the curve with equation $y = 3e^x + 2x$, find the value of x for which the gradient of the tangent is 5.

- ③ For the curve with equation $y = 2\log_e x$, find

- A the gradient of the tangent to the curve at the point $(e, 2)$

- B and hence the equation of the tangent to the curve at the point $(e, 2)$.

- ④ For $f: R \rightarrow R$ where $f(x) = x^3(x - 3)$, find the equation of the tangent at the point where $x = 2$.

- ⑤ Find the gradient and hence the angle the graph with equation $y = \sin 4x$ makes with the positive x -direction at the point where $x = \frac{\pi}{2}$. Give the angle in decimal degrees correct to 2 decimal places.

- ⑥ A Find the gradients of the tangent and normal to the curve with equation $y = \sin 2x$ at the point $(\frac{\pi}{6}, \frac{\sqrt{3}}{2})$.

- B What are the angles that the tangent and normal make with the positive x -direction?

THINGS TO KNOW

Know the method!

$f'(x)$ gives the gradient of the tangent to the curve with equation $y = f(x)$ at any point x . The equation of the tangent to this curve at the point (x_1, y_1) can be found by using the formula $\frac{y - y_1}{x - x_1} = m$ where $m = f'(x)$.

HINT

Read the question!

Yes, it says 'find the equation' so get the equation, not just the gradient. It's probably easier to multiply out the brackets first.

HINT

Remember that the angle a line with gradient m makes with the positive x -direction can be found using $m = \tan \theta$.

THINGS TO KNOW

The equation of the normal to the curve $y = f(x)$ at the point (x_1, y_1) can be found by using the formula $\frac{y - y_1}{x - x_1} = m$ where $m = \frac{-1}{f'(x)}$.



16 – Applications of differentiation to tangents to curves (cont'd)

- ③ Find the equations of the tangent and normal to this curve at $(\frac{\pi}{6}, \frac{\sqrt{3}}{2})$.

- ⑦ The graphs with equations $y = e^{x^2} - 2$ and $y = 2 - e^{x^2}$ intersect at two points. Find those points.

HINT

Careful of the simultaneous equations and handling the square root of $\ln 2$. Keep your answers exact.

- ⑧ Two graphs intersect at a point. Find the angle in degrees between the two graphs if

- Ⓐ the gradient of the first graph is $\frac{1}{2}$ and of the second is -2 .

- Ⓑ the gradient of the first graph is $\frac{1}{2}$ and of the second is $\frac{1}{4}$.

- ⑨ The graphs with equations $y = f(x)$ and $y = g(x)$ intersect at two points. If $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$

- Ⓐ show that the graphs intersect at the points $(2, 0)$ and $(-2, 0)$.

- Ⓑ find the gradient of the tangent to the graph $y = f(x)$ at the point $(2, 0)$.

- Ⓒ find the angle that the tangent to the graph $y = f(x)$ makes with the x -axis at the point $(2, 0)$ in decimal degrees correct to two decimal places.

- Ⓓ find the gradient of the tangent to the graph $y = g(x)$ at the point $(2, 0)$.

- Ⓔ find the angle that the tangent to the graph $y = g(x)$ makes with the x -axis at the point $(2, 0)$ in decimal degrees correct to two decimal places.

- Ⓕ hence find the angle between the two graphs at the point $(2, 0)$.

THINGS TO KNOW

Two graphs with gradients m_1 and m_2 are perpendicular if $m_1 m_2 = -1$.

The angle $\alpha = \theta_2 - \theta_1$ between two graphs with gradients m_1 and m_2 is given by the formula:

$$\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

TRAP

Be careful of rounding too early.

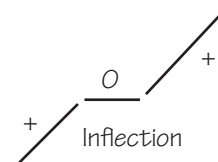
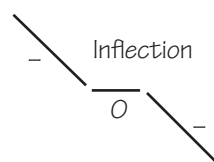
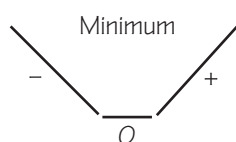
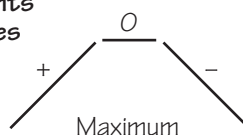
17 – Graphs using calculus

Stationary points are points where the gradient of the graph is zero and these will occur at maxima, minima or horizontal points of inflection. It is important to be able to identify the nature of the stationary points and you can do this using calculus [$f''(x)$] or by looking at the gradient [$f'(x)$] on either side to see if it is positive or negative.



THINGS TO KNOW

Gradients of curves



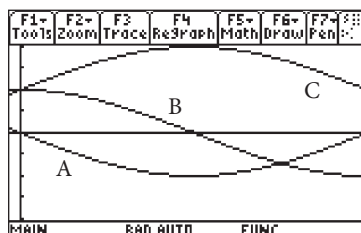
- ① Use calculus to find the stationary points of the function $h(t) = 2t(t^2 - 3)$. Identify each stationary point (maximum, minimum or inflection point).

HINT

'Use calculus', but of course you can use the calculator to check your answer.

- ② The graphs with equation $y = f(x)$, $y = f'(x)$ and $y = f''(x)$ are shown. Match the correct equation to its graph.

- A _____
B _____
C _____



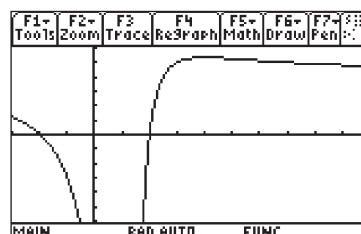
HINT

You don't know the rule of f but you're finding turning points where $f'(x) = 0$ so you don't need the rule. Your calculator is probably of little help here!

- ③ For the graph of $y = f(x)$ shown, the x -intercepts are at $(-2, 0)$ and $(2, 0)$ and the y -intercept is at $(0, -16)$. There is an asymptote at $x = 1$ and $y = 1$ and a maximum at $(4, 5\frac{1}{3})$.

- A Complete the following table.

$f(x)$	$f'(x)$ positive, negative or zero	Values of x
increasing		
decreasing		
zero		
maximum		
minimum		
	zero	



- B Use the table to draw a rough sketch of the graph of $y = f'(x)$.

- ④ A Expand $f(x) = (x + 5)(x - 4)^2$.

TRAP

- Be careful of algebraic errors when you are multiplying out.
- if you use $f''(x)$ to test for stationary points, remember that this may not be enough for inflection points.
- Concave down means $\cap$ -shaped.



- Ⓑ If the turning points of the graph of $y = f(x)$ are at $x = 4$ and $x = -2$, find the coordinates of these turning points.

- Ⓒ Test and hence state the nature of the turning points of the graph of $y = f(x)$.

- Ⓓ Find the values of x for which the graph is concave down.

HINT

A calculator is no help – you need to understand the theory.

- 5 The graph of the function $f: \mathbb{R} \setminus \{-2\} \rightarrow \mathbb{R}$ has a vertical asymptote with equation $x = -2$ and a horizontal asymptote with equation $y = 1$. It also has the following properties.

- $f(0) = 2$
- $f'(x) < 0$ for all $x < -2$
- $f'(x) < 0$ for all $x > -2$

Draw a sketch of a possible graph of $y = f(x)$.

- 6 For a function $f(x)$ over the set of real numbers, $f'(x) = 3x(x - 2)$ and the graph of $y = f(x)$ has turning points $(u, 5)$ and $(v, 1)$. Find the values of u and v .

HINT

Intercepts should be labelled as number pairs, asymptotes as equations. This graph has two branches and you need to draw both. Be careful that the graph is not shown curving away from the asymptotes, or crossing the asymptotes.

- 7 Ⓐ For the function $f: \mathbb{R} \setminus 1 \rightarrow \mathbb{R}$ where $f(x) = 2 - \frac{2}{x+1}$, sketch the graph of $y = f(x)$, labelling all intercepts and asymptotes.

- Ⓑ Use your graph to state values of x for which

- $f(x) = 0$. _____
- $f'(x) > 0$. _____

- 8 Ⓐ The equation $x^3 - 3x^2 - 9x + 11 = 0$ can be written as $(x - 1)(x^2 - 2x - 11) = 0$. Find the exact solutions to this equation.

- Ⓑ If $f(x) = x^3 - 3x^2 - 9x + 11$, find $f'(x)$ and hence find the nature and coordinates of the turning points of the graph of $y = f(x)$.

- Ⓒ On a separate sheet of paper, sketch the graph of $y = f(x)$, showing the coordinates of the intercepts with the axes and the turning points.

18 – Maximum and minimum tricks and traps

There are always clues in any question on maximum and minimum, and you should learn to recognise them.



TRICK

Certainly you should check to see whether the question is wanting the maximum/minimum value or the value of the variable for which it occurs. You will usually need to differentiate a function if it is a local maximum/minimum. You will need to confirm that you have a maximum or minimum or just an inflection point. (Do this by either checking the gradient on either side of the point or using the second derivative if that's what you've been taught.) If the question asks for an absolute maximum or minimum, this is a clue to check the end points of the domain. Whatever you do, always sketch a graph if you can. And if you are asked to show something, please don't fiddle with the algebra to try and get the answer given; your working gets the marks, not fiddled working!

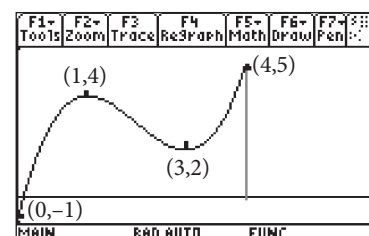
- ① For the function with the graph shown on the domain $0 \leq x \leq 4$, state:

- Ⓐ the local minimum value of the function and the value of x for which the local minimum occurs.

- Ⓑ the local maximum value of the function and the value of x for which the local maximum occurs.

- Ⓒ the absolute minimum value of the function within the domain and the value of x for which the minimum occurs.

- Ⓓ the absolute maximum value of the function and the value of x for which the maximum occurs.



HINT

If the question asks for an absolute maximum or minimum, this is a clue to check the end points of the domain.

- ② A real function has gradient function $f'(x) = 4x^3 - 16x$ and the graph has maxima, minima or inflection points at $(-2, -9)$, $(2, -9)$ and $(0, 7)$. Use $f''(x)$ or otherwise to identify the nature of the each of the turning points.

- ③ For a function $f(x)$ where $f'(2) = 0$, $f'(1) = -3$, $f'(3) = 5$, what sort of turning point is at $x = 2$?

- ④ Find the minimum value of y given that $y = xe^x$, $x \in \mathbb{R}$.

HINT

Don't just find x .

- ⑤ The curve with equation $y = ax^2 + bx$ has a maximum value at the point $(\frac{1}{3}, \frac{1}{3})$. Find the values of a and b .

TRAP

Careful! Maybe you don't have to differentiate.



18 – Maximum and minimum tricks and traps (cont'd)

- 6 For the function $f(x) = \frac{1}{6}x^3 - 2x^2 + x - 2$, find

(A) the gradient function $f'(x)$.

(B) the value of x for which the gradient of the curve with equation $y = f(x)$ is a maximum.

- 7 For what value of x does the function $f(x)$ have a maximum value if the graph of its gradient function is as shown?

- 8 The probability of 'getting two greens' in an experiment is given by $6p^2(1-p)^2$. Find p so that the probability of 'getting two greens' is a maximum. Explain your answer.

- 9 A function, f , has rule $f(x) = \frac{1}{x+1} + \frac{4}{8-x}$, $0 \leq x \leq 7$.

(A) Find the value of x for which the function is a minimum, and find the value of the function at that point.

(B) On the graph shown, what are the coordinates of the points labelled A, B and C?

(C) Find the value of x for which the function is a maximum within the domain, and find the value of the function at that point.

- 10 The profit, P dollars, made on the production and sale of x Queen-sized doonas can be approximated by the function $P(x) = 900 \log_e(1 + \frac{x}{300}) - 200 - x$.

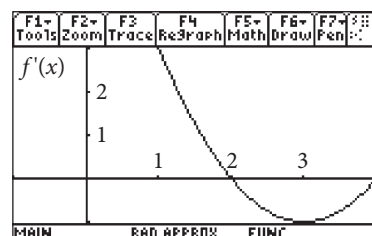
(A) Find the number of doonas that should be produced and sold for maximum profit.

(B) Calculate this maximum profit. _____

(C) Use the graph of P shown to estimate the minimum and maximum numbers of doonas that can be produced and sold so that a profit will be made.

HINT

Read the question carefully.
What is a maximum?

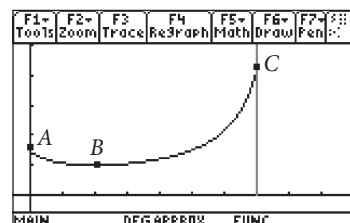


HINT

Expand rather than using the product rule.
What is the domain of p ?

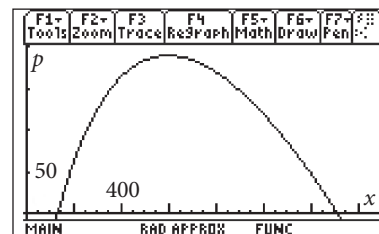
TRAP

Careful with the signs when you differentiate the second fraction and watch your algebra! The words 'within the domain' and the graph you have looked at are a clue for Part (c).



TRAP

Be careful with the differentiation and make sure you read the question.



19 – Maximum and minimum ‘wordies’

Practical problems to find maximum or minimum values are easily identified because they usually have words like ‘greatest’, ‘least’, ‘most’, ‘closest’ as well as the words ‘maximum’ or ‘minimum’.



THINGS TO KNOW

READ

- ☞ What is to be maximised/minimised; say, f ? (If the question asks for an absolute maximum or minimum, this is a clue to check the end points of the domain.)
- ☞ What is varying? Perhaps there are two things.

PREPARE

- ☞ Do you have a rule for the function f to be maximised/minimised or can you find it? Perhaps you have been asked to show this so even if you can't do the ‘show’, you can still use it later.
- ☞ If there are two variables, you need to find a relationship between them and then get rid of one of the variables in the rule for f . Perhaps you have been asked to show this so make sure your explanations are clear (and don't try to fudge the answer).

DO

- ☞ Differentiate f and put equal to zero for maximum/minimum/inflection.
- ☞ Solve this equation to give the value of the variable for which the derivative is zero.

CHECK

- ☞ You need to confirm that you have a maximum or minimum or just an inflection point. Do this by either checking the gradient on either side of the point or using the second derivative if that's what you've been taught. Whatever you do, it will be helpful to sketch a graph if you can.
- ☞ Answer the question asked: does the question ask for the maximum/minimum value or just the value of the variable for which it occurs?
- ☞ Check that your answer is feasible and sensible.

REREAD TO CHECK YOU'VE FINISHED THE QUESTION

- ☞ Tick off each part of the question to make sure you have answered it.

READ

- ① The area of the sector of a circle of radius r cm with angle θ radians shown is 10 cm^2 .

PREPARE

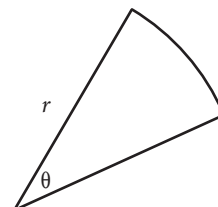
- Ⓐ Use the formula for the area of a sector, $A = \left(\frac{1}{2}\right)r^2\theta$ to show that $\theta = \frac{20}{r^2}$.

- Ⓑ If P is the perimeter of the sector, use the formula for the length of an arc of a sector $l = r\theta$ to write an expression for P .

- Ⓒ Hence write an expression for P in terms of r only.

DO

- Ⓓ Find the **exact** value of r for which P is a minimum, and find the minimum value of P .



HINT

Note the words ‘exact’ and ‘hence’. You should check that the answer is physically feasible.

CHECK

- Ⓔ Confirm that the value is a minimum.

REREAD TO CHECK YOU'VE FINISHED THE QUESTION



- ② A window 1.5 m high is placed in a wall with its window sill 2 m above your eye level. You are standing x metres from the wall.

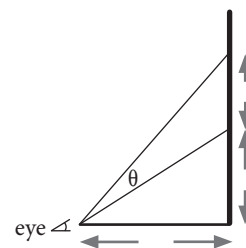
(A) Label the diagram.

- (B) You wish to know how far you should stand from the wall so that the angle θ subtended at your eye by the window will be maximum. You find that the expression connecting θ and x is $\tan\theta = \frac{1.5x}{x^2+7}$ and that maximum θ occurs when $x = \sqrt{7}$.

What is the value of the maximum angle, correct to two decimal places?

HINT

Worded questions can be hard to understand, but the examiners are usually kind enough to help you through.



HINT

Be careful to answer the question asked and check that your calculator mode is in degrees, not radians.

- ③ How much do you learn during a lesson? Let x units be the amount of information your teacher gives out during the lesson and let y units be the amount of information you take in during the lesson.

Suppose that $y = xe^{-0.05x}$ where $x \geq 0$.

(A) Explain the restriction on the values of x .

- (C) Find how much information the teacher should give out during a lesson so as to maximise the amount of information you take in.

HINT

Product rule is needed for differentiation. The question asks for proportion, which means amount learned as a fraction of what was given by the teacher.

- (B) Find $\frac{dy}{dx}$.

- (D) If the teacher gives this amount of information, what proportion of this information do you think you will take in? Give your answer correct to 3 decimal places.

- ④ A machine moves up and down according to the rule $y = 30 + 25 \sin\left(\frac{(4t-1)\pi}{2}\right)$; $t \geq 0$. The distance of the machine from the ground is y metres at time t minutes from the start.

(A) What is the maximum value of y ? _____

(B) What is the minimum value of y ? _____

(C) After how long from the start is the first minimum? _____

(D) After how long does the machine reach a height of 40 metres from the ground? Give your answer in seconds, correct to two decimal places.

(E) For how long is the machine more than 40 metres from the ground during the first minute? Give your answer correct to 3 decimal places.

HINT

You don't always need to use calculus! Look at the function and its values. Make sure you are in radians and that your answers do not have π in them. Remember, the table on your calculator will give whole numbers.

20 – Formulae in maximum and minimum ‘wordies’

Worded problems often come out of practical situations where you need to remember your mensuration formula like area, volume and perimeter. Some of these will be given on your exam formula sheet.



THINGS TO KNOW

Circumference of a circle: $2\pi r$

Arc length: $r\theta$ (θ in radians)

Area of a square: l^2

Area of a rectangle: $l \times b$

Area of a trapezium: $\frac{1}{2}(a + b)h$

Area of a circle: πr^2

Area of a sector: $\frac{1}{2}r^2\theta$

Area of a segment: $\frac{1}{2}r^2(\theta - \sin \theta)$

Area of a triangle: $\frac{1}{2}bc \sin A$

Surface area of a cylinder (including top and bottom): $2\pi r^2 + 2\pi rh$

Surface area of a cone (including base): $\pi r^2 + 2\pi r\ell$ (ℓ is the slant height)

Surface area of a sphere: $4\pi r^2$

Volume of a cuboid: $l \times b \times h$

Volume of a cylinder: $\pi r^2 h$

Volume of a cone: $\frac{1}{3}\pi r^2 h$

Volume of a pyramid: $\frac{1}{3}Ah$

Volume of a sphere: $\frac{4}{3}\pi r^3$

- ① Rewrite the above formulae for lengths, areas and volumes and draw appropriate diagrams showing the meaning of each dimension.
- ② An open container is to be made from a rectangular sheet 12 cm by 24 cm by cutting equal-sized squares of side-length x cm out of the four corners and turning up the sides.

Ⓐ What is the height of the container made?

Ⓑ Show that the volume, V cm³, of the container is given by $V(x) = 4x^3 - 72x^2 + 288x$.

Ⓒ What is the domain of the volume function, V ?

Ⓓ Find the exact value of the maximum volume of the container.

- ③ The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of side lengths $2r$ metres and s metres.

Ⓐ Find an expression for the perimeter, P , of the pool in terms of r and s .

Ⓑ If the surface area of the pool is constant at A m², find an expression for the area in terms of r and s .

Ⓒ Hence find an expression for

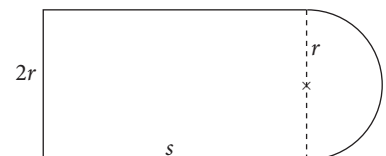
• s in terms of r . _____

• P in terms of r and A . _____

TRAP

Even if you can't do the 'show', you can still use its results.

Be careful because you are asked for the exact value, not a decimal, and you are finding the maximum volume not just the value of x .





20 – Formulae in maximum and minimum ‘wordies’ (cont’d)

- ④ Show that the perimeter P of the pool is minimum when r and s are both equal to $\sqrt{\frac{2A}{4+\pi}}$.

HINT

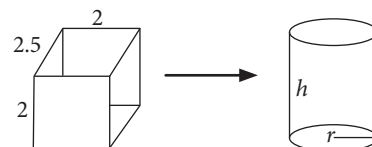
Don't forget s !

HINT

Remember that A must be treated as a constant (like a number). You should also confirm that the answer you got really was a minimum.

- ⑤ Find the minimum perimeter of the pool.

- ④ A block of metal in the shape of a rectangular prism with dimensions 2 cm by 2 cm by 2.5 cm is melted down and then recast into a solid cylinder with base radius r cm and height h cm.



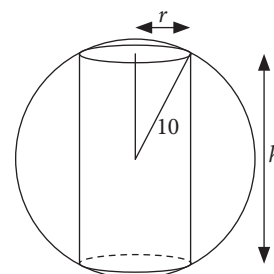
- ① Find the volume of metal in the cast cylinder in cm^3 and hence express h in terms of r .

- ② The surface area of a cylinder is given by $A = 2\pi rh + 2\pi r^2$. Show that the surface area of the cast cylinder is given by $A = \frac{20}{r} + 2\pi r^2$; $r > 0$.

- ③ Find the value of the radius (correct to 2 decimal places) for which the surface area of the cylinder is a minimum.

- ④ What is the height of the cylinder (correct to 2 decimal places) for minimum surface area of the cylinder?

- ⑤ A hole is drilled through the centre of a solid metal sphere of radius 10 cm as shown. The diagram shows a cross-section of the sphere and the hole. The radius of the cylindrical hole is r cm and h cm is its height.



- ① Use Pythagoras' Theorem to express r in terms of h .

- ② Hence show that the volume, $V \text{ cm}^3$, of the cylindrical hole is given by $\frac{\pi h}{4} (400 - h^2)$.

- ③ Find the maximum value of V , giving your answer correct to one decimal place.

HINT

- ☞ Use the methods suggested and note the 'hence'.
- ☞ Even if you can't do the 'show', you can still use the result.
- ☞ Be careful of the algebra!
- ☞ Multiply out brackets before differentiating.
- ☞ Test for maximum.

21 – Checklist for success

Everyone works differently so you should change this checklist to suit yourself and add some other things if you like. The important thing is to have a plan. That way you are less likely to put off the things you find hard. Finally, don't stress about a problem, do something about it: spend more time on it or see the teacher.



Revision time	Done ✓
I have tried all my homework and asked the teacher about any problem areas.	
I have made a revision plan by dividing the course into blocks of time and have worked hard to stick to it.	
My revision plan includes recreation and relaxation.	
My revision plan includes a couple of free times in case I need to catch up because of sickness, etc.	
I have altered my revision plan (sickness, etc.) immediately when I needed to and not waited until the end.	
I have helped myself learn by making my own notes of important points with examples, and organising a summary of the work in my own words.	
I have thought about key ideas and methods, learned formulae, made summary tables and/or mind maps.	
I have ticked off my list of things as I have revised and always look at how much I've done rather than how much I have left to do.	
I have rewarded myself each time I reached a target by going to the refrigerator or watching TV or calling a friend.	
I have moved to the next topic knowing that I can talk to myself about the definitions of terms, the key points, real examples, formulae and graphs.	
I know what sorts of questions I am likely to be asked: short answer, multiple choice, ...	
I have practised each type of exam question: theory, skills, interpret, analyse, ...	
I know what to do with questions with several parts.	
I have worked through at least the last two year's exams against the clock in the required time using only the tools I am allowed to use in the real exam.	
I have had practice working for 2–3 hours without a break like I will need to do in the exam.	

I have worked out how long each type of question should take by dividing the total exam time for each type by the number of questions.	
I have checked that I have an approved calculator and spare batteries.	
I know what will be on the formula sheet provided.	
I have checked that I have pencils, pens, highlighters, ruler, eraser and other mathematical equipment.	
I know what written notes (if allowed) and other things I can take into the exam.	
I have a watch or clock I can use.	
I have checked when the exam is on and where.	
I have worked out how long it takes to travel to the exam and how I will travel.	
The night before the Exam	Done ✓
I have put together what I need for the exams: calculator, spare batteries, pencils, pens, highlighters, written notes (if allowed), ruler, eraser and other mathematical equipment, and a watch or clock.	
I have put out the clothes I need.	
I have not done any more questions or tried to 'cram', but have just read through my notes to refresh my memory and looked at questions I have already done correctly.	
I have checked the length of the paper and worked out how long I should spend on each question.	
I keep telling myself how much I know and understand.	
I have double-checked when the exam is and where.	
I have double-checked my alarm clock (mechanical and human) to give myself plenty of time.	



I have had a shower or a bath to relax and gone to bed at a reasonable time.	
Exam Morning	<i>I will</i> ✓
Have a good breakfast or lunch before I leave home.	
Collect all my exam equipment.	
Leave home in plenty of time to get there.	
Check that I am at the right room and the starting time for the exam.	
During the Exam – Good Habits	<i>I will</i> ✓
Use my reading time to read the paper carefully, set times for different questions/sections and to work out which questions I will do first and which ones I think are harder and need to be done later in the exam.	
Spend the correct amount of time on each question and go on to the next one even if I haven't quite finished because I know I will have most of the marks for the answer by then anyway.	
Remember that 1 mark means a mark for the answer only and more marks mean more work.	
Return to unfinished questions after I have attempted all the questions I need to.	
Try to write something down if I have a mental block because I know that writing down a few relevant facts, equations or formulae, will probably jog my memory enough to get started (and I won't get any marks for a blank sheet of paper).	
Never leave out answers to multiple choice or short answer questions. I will have a go since there are no marks lost for wrong answers and I won't get marks for gaps!	
Always try to ask myself 'What does the examiner want?'	

Answer the question asked, not the one I think was asked.	
Use the method the examiners have said to use (if they have).	
Give the answer in the form the examiners have asked for (if they have).	
Reread the question when I have finished to make sure I have answered the question.	
Be extra careful with questions that I find easy because I may not think carefully enough.	
Watch out for key words like 'hence' (use something done or given in an earlier part) and 'show' (prove or demonstrate).	
Not use the calculator when the question asks for exact answers.	
Use my calculator wisely, making sure that I put in brackets where I need them, that I am in the right mode (radian/degree).	
If I have a graphical calculator, always adjust the window so I can see the key features of any graph.	
If I have a graphical calculator, use the CALC and not the TRACE button.	
Take care to avoid making arithmetic and algebraic mistakes.	
Show all my working.	
Make sure my answers are written legibly and logically.	
Above all, take the chance I have been given to show the examiner what I know and understand and will feel confident that I have done my best, whatever my mark.	

Answers

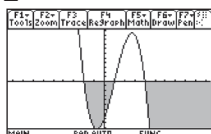
Note – No answers are provided to questions such as 'show' and 'prove'.

1 – Tricks and traps in arithmetic

- 1** 179 324 873 046 **2** **A** 1 **B** 7 **3** $5/24$
4 **A** yes **B** 4 or 6 **C** no **D** no
5 **A** 14.47 so 14 **B** -0.50
6 4.5267...; one is 4.5 and the other is 4.53
7 $6 \times 6\,000 - 40\,000 < 0$; 2590.74!; second-digit rounding
8 **A** 3.7320... **B** 8.1681... **C** 2.1886...; 2.19 to 2 d.p.
9 0 **10** -437.5 **11** 5.88×10^{18} **12** 2

2 – Tricks and traps in algebra

- 1** **A** probably 0 **B** yes **C** $x = -2$ because it makes the denominator zero
2 $y = \pm\sqrt{9-x^2}$
3 **A** 3, -2 **B** no, -2 gives LHS = 2 and RHS = -2
4 $x^3 - 64 = (x-4)(x^2 + 4x + 16)$
5 $(3 \times 2 + 1) + (3 \times 3 + 1) + (3 \times 4 + 1) + (3 \times 5 + 1) = 46$
6 **A** $(x+y)^2 = x^2 + 2xy + y^2$ **B** $a^2 - 2ab + b^2$ **C** $(a-b)^2$
7 $16x^2 - 128x + 256$ **B** **A** $x < -2$ and $1 < x < 4$
B



- 9** **A** $3x^2 + 3xh + h^2 + 1$ **B** $3x^2 + 1$
10 **A** $-1 \leq x - 2 \leq 1$ to give $1 \leq x \leq 3$
B $+(x-2) \leq 1$ or $-(x-2) \leq 1$ to give $x \leq 3$ or $x \geq 1$
C 3, 1; graph to give $x \leq 3$ or $x \geq 1$
D Probably $-1 \leq x - 2 \leq 1$ method

3 – Expanding and factorising – best way

- 1** $27a^3 - 27a^2 + 9a - 1$ **2** $-108xy^6$ **3** $2(y^2 - y + 3)$ with remainder -19
4 **A** $2(x+2y-9z)$ **B** $4(2a-b)(2a+b)$ **C** $(z-\sqrt{5})(z+\sqrt{5})$
D $3(x+2y)(x^2 - 2xy + 4y^2)$ **E** no real factors
5 **A** $(3y-2)^2$ **B** $(2x-3)(x+2)$ **C** $(x-3-\sqrt{5})(x-3+\sqrt{5})$
D $(x-\sqrt{3})(x+\sqrt{3})(x-\sqrt{2})(x+\sqrt{2})$ **E** $(x+2)(x^2 - 2x + 6)$
6 Yes; discriminant perfect square; rational (fractional) factors
7 **A** $(x-2)(x^2 + 2x + 4)$ **B** $2\left(x - \frac{5}{4} - \frac{\sqrt{57}}{4}\right)\left(x - \frac{5}{4} + \frac{\sqrt{57}}{4}\right)$
C $(x+3y+2z)(x+3y-2z)$ **D** $(x+4)(x-5)(x+2)$
8 $(y-1)(2y^2 + 13y + 15)$; $(y-1)(2y+3)(y+5)$
9 $x(\sqrt{a}x - \sqrt{b})(\sqrt{a}x + \sqrt{b})$

4 – Indices and logs

- 1** 9; 2 **2** 5x **3** -1.532 **4** 9
5 **A** $k = \log_e 4$ for all $x \in \mathbb{R}$ **B** $k + kx$
6 140 **7** No, too small. **8** Take log of both sides
9 **A** 1.63 **B** 1.6 **10** $k = \log_e 2$
11 **A** $e^x(e^x + e^{2x})$ **B** $(e^x - 4)(e^x - 2)$ **12** greater than
13 $x = \frac{1}{k} \log_e \frac{3}{4}$ **14** $x < -\log_e 0.5$ or $x < 0.6931$

- 15** **A** $-1 \leq x \leq 1$ excluding $x = 0$ **B** $0 < x \leq 1$ **C** $x > 0$

5 – Solving equations and inequations

- 1** -2 **2** i $-4 \leq x \leq 4$ ii $-4 \leq x \leq 4$ **3** $x > 5$ and $x < 1$ **4** $-2, \frac{4}{3}$
5 -2, 2 **6** 2 real solutions **7** $\pm\sqrt{2}$ **8** (2, 5) and $\left(-\frac{1}{2}, \frac{5}{4}\right)$
9 $y = (x+1)(x^2 + 4x + 4)$; -1, -2 **10** **A** 4, -4 **B** -1
11 $-3 < y < 0$ **12** $2 \leq x \leq 12$ **13** $bc - a$

6 – Exponential and logarithmic functions

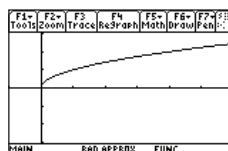
- 1** $a > 0$, $a \neq 1$ or $a \in \mathbb{R}^+ \setminus \{1\}$; $N > 0$,
2 **A** y -axis is an asymptote; draw this asymptote clearly
B no y -intercept; x -intercepts at $(-1, 0)$ and $(1, 0)$
3 **A** $x > 1$; R **B** $f^{-1}(x) = e^{\frac{1}{2}(x-3)}$ **C** R ; $y > 1$
5 **A** domain: $(2, \infty)$ or $2 < x < \infty$; range: R **B** 1:1
C $f^{-1}(x) = 2 + e^{5x}$
6 0.25

7 – Growth and decay

- 1** **A** $k = 0.039925 \dots \approx 0.04$, $Y_0 = -30$ **B** 37.08
2 **A** $t = 50 \log_e \left(\frac{P}{4}\right)$ **B** 35 **3** 0.0958
4 **A** 10 **B** 0.1 (units are thousand million)
5 **A** $s \geq 0$ **B** $\frac{22}{3} = 7\frac{1}{3}$ **C** 2.465
6 **A** I when $t = 0$ **B** a value of I at a given value of t **C** 30
7 **A** 1.155 minutes **B** 2 200 **C** 195.080
D 91.133% killed (8.867% left) **E** 0.912 minutes
8 **A** unlikely; an exponential increases at an increasing rate
B there was a pause in sales at about 3.75 years
C there is a limit (7) to the number of rabbits that can be housed; it can't equal 8 if it is an asymptote and so must be 7 since it needs to be a whole number of rabbits.

8 – Functions of functions of functions of...

- 1** $f^{-1}(x) = \frac{x+7}{2}$ **2** $y = 2$
3 **A** $f^{-1}(x) = \pm\sqrt{x}$ **B** $f(x)$ is not 1:1
C either $x \geq 0$ or $x \leq 0$ or similar
D

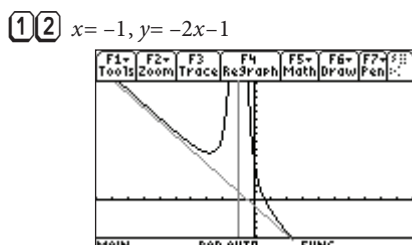
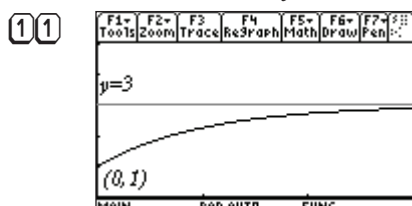


- 4** $f(x-1) = \frac{23-2x^2}{7(x+1)}$
5 **A** $\frac{1}{x^2(x-2)}$ **B** $\frac{1-2x}{x^3}$ **C** $\frac{1-2x}{x^3}$ **D** $\frac{1}{x^2(x-2)}$ **E** $y^3 - 2y^2 = x$
6 **A** (a) $f[g(x)] = 3x^2$ **B** $g[f(x)] = 9x^2$ **C** Yes
7 **A** $6x^2 - 3x$ **B** $18x^2 - 51x + 36$ **C** $-12x^2 + 48x - 36$
D $216x^2 - 594x + 408$
8 1:1 **9** Not 1:1
10 **A** is not; not 1:1 **B** $x \geq -2$ or $x \leq -2$
C $f^{-1}(x) = \sqrt{x} - 2$ for $x \geq -2$ or $f^{-1}(x) = -\sqrt{x} - 2$ for $x \leq -2$
11 (-2, -2)
12 **A** T **B** T **C** F **D** F **E** F **F** T
13 $0 \leq t \leq 2\pi$; the intersection of the domains of y_1 and y_2



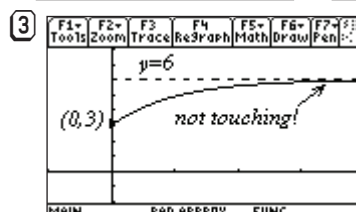
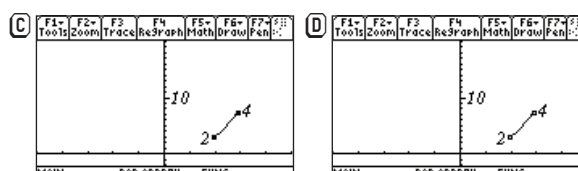
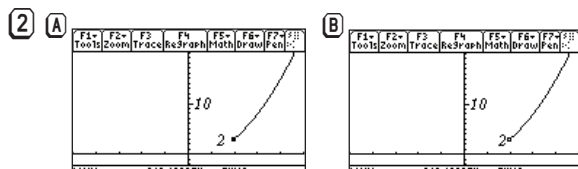
9 – Transformation of graphs

- 1 (A) $y = 3(x-2)^2$ (B) $y = 3x^2 + 2$
 2 $\frac{1}{3}; 2$
 3 Dilation of factor of 3 from the y -axis (in the x -direction) and translation 1 unit in the y -direction.
 4 (A) Dilation factor 2 in y -direction, reflection in x -axis, translation 1 unit in negative x -direction and 2 units in negative y -direction
 (B) Translation 2 units in negative x -direction and 5 units in negative y -direction (C) Dilation factor 2 in y -direction, dilation factor $\frac{1}{2}$ in x -direction, translation π units in x -direction
 (D) Dilation factor 3 in y -direction, dilation factor $\frac{1}{2}$ in x -direction, reflection in the X -axis, translation 2 units in x -direction
 5 (A) $y = (2x+4)^3$ (B) $y = 8(x+4)^3$
 6 dilation factor 2 parallel to the y -axis, dilation factor $\frac{1}{3}$ parallel to the x -axis; translation $\frac{\pi}{3}$ in the x -direction; translation 1 in the y -direction
 7 (A) $x = 1; y = 4$ (B) $y = 1$ (C) $x = -\frac{2}{3}$
 8 $g(x) = (x-2)^3; h(x) = (x-2)^3 - 1; j(x) = (3x-2)^3 - 1$
 9 (0, 3); (6, -1)
 10 (A) $y = e^{x+3}$; translation of graph of $y = e^x$ 3 units in the negative x -direction (B) $y = \log_e(3x)$; dilation factor $\frac{1}{3}$ in x -direction

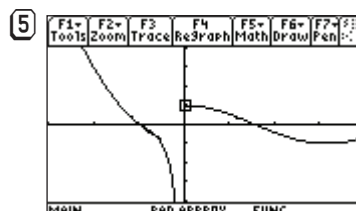


10 – Intervals and restricted graphs

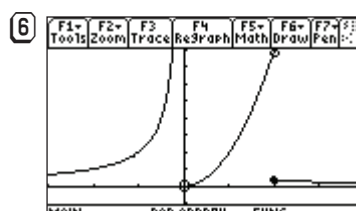
- 1 (A) $x \geq -1$ or $[-1, \infty)$ (B) R (C) $0 < x \leq 2$ or $(0, 2]$



- 4 (A) $y = |x|$; no; yes, not differentiable
 (B) $y = 2$ for $x > 0$ and $y = -2$ for $x < 0$; no; no, not differentiable
 (C) $y = 2$ for $x > 0$ and $y = -2$ for $x \leq 0$; no; no, not differentiable
 (D) $y = x^2 - 1$; yes; yes, differentiable

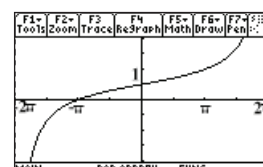


Domain is R excluding $\{0\}$, range is R



R excluding $\{0\}$

- 7 asymptotes at $x = \pm 2\pi$



11 – Trigonometric functions and graphs

- 1 $R = D \frac{\pi}{180^\circ}$ because $\theta^\circ = \left(\theta \times \frac{\pi}{180^\circ}\right)^\circ$

- 2 (A) $\frac{5\pi^c}{4}$ (B) $\frac{11\pi^c}{6}$

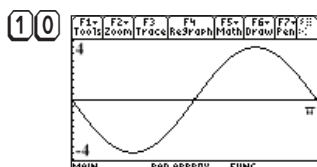
- 3 (A) -0.741 (B) 1.816

- 4 (A) 1.819 (B) -0.7568

- 5

Special Angles	0° 0^c	30° $\frac{\pi^c}{6}$	45° $\frac{\pi^c}{4}$	60° $\frac{\pi^c}{3}$	90° $\frac{\pi^c}{2}$
Sine	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
Cosine	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
Tangent	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef

- 6 (A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) $-\frac{\sqrt{3}}{2}$ 7 $2 + \sqrt{3}$ 8 $\frac{\pi}{4}, \frac{3\pi}{4}$ 9 12π



11 A 23 B 1 a.m. the next day

12 – Polynomial functions and their graphs

1 $y = -\frac{2}{3}x + 2$

2 $y = 2(x-1)^2 + 1$ multiplies out to $y = 2x^2 - 4x + 3$; (1, 1); 3

3 $x = 1$ and $x = -\frac{2}{3}$

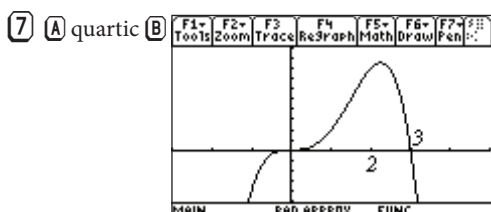
4 $y = -x^2(x+1)(x-3)$ but could have other answers like $y = -2x^2(x+1)(x-3)$ or $y = -\frac{1}{2}x^2(x+1)(x-3)$, etc.

5 $f(x) = (x-b)^2(x-c)$ but could have other answers

6 A $d = 10$; $125a + 25b + 5c + d = 0$;

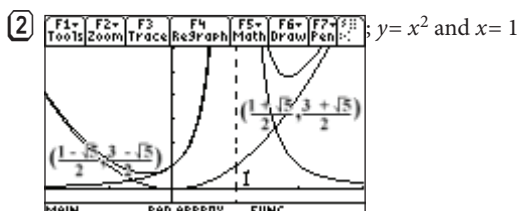
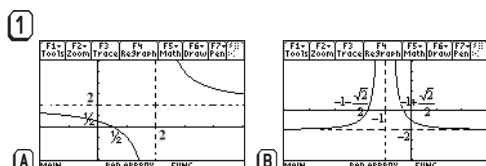
$a + b + c + d = -12$; $-8a + 4b - 2c + d = 0$

B $f(x) = 2x^3 - 7x^2 - 17x + 10 = (2x-1)(x+2)(x-5)$

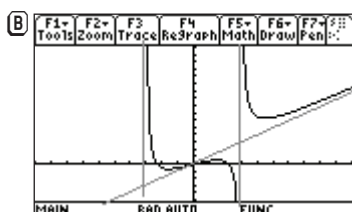


C $p = -1$; $q = 3$

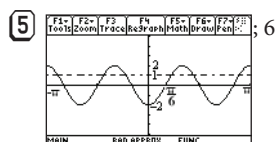
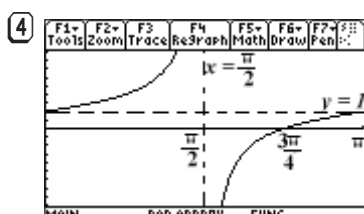
13 – Other graphing practice



3 A asymptotes $x = +3$, $x = -3$, $y = x$



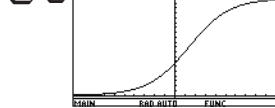
C $(-\frac{1}{4}, -\frac{1}{4})$



6 $2\cos(10\pi t)$ 7 $0 \leq k \leq 5$

8 A $-1 < x < 2$ and $x > 2$ B $(x+h-2)^2(x+h+1)$ C $-1 < h < 2$

9 A asymptotes: $y = 0$, $y = 10$



C $(-\log_e(\frac{1}{2}), 5)$, $(\log_e(\frac{1}{2}), 5)$

14 – Tricks and traps in differentiation

2 A $2x - 9x^2 + 17$ B $15e^{15x}$ C $5\cos 5x$

3 A product B chain rule (function of a function) C quotient

4 A $3(3x^4 - 4)^2(12x^3)$ B $4\left(\frac{1}{x} + 2x\right)^3\left(-\frac{1}{x^2} + 2\right)$

C $\frac{\ln(x+1)(2x) - (x^2+2)(\frac{1}{x})}{(\ln x+1)^2}$ D $\left(\frac{1}{2}x^{-\frac{1}{2}}\right)[\ln(3x)+2x] + (\sqrt{x}+2)\left(\frac{1}{x}+2\right)$

E $\frac{(6.5e^{0.3t-15})(0.1e^{0.02t}) - (5e^{0.02t})(1.95e^{0.3t-15})}{(6.5e^{0.3t-15})^2}$

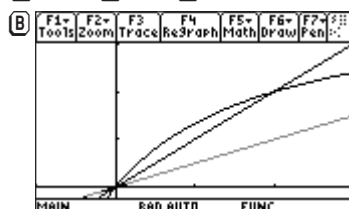
5 A $2x\cos 5x - 5x^2 \sin 5x$ B $5(m-1)^7(9m-1)$ C $2(6x+1)e^{3x^2+x}$

D $\frac{3}{3t-1}$ E $\frac{-12x}{(3x^2+1)^2}$

6 D $f'(e^{2x})$ E $2f(x)$ 7 $f'(x) = 3x(x-2)$ so $a = 3$ and $b = 0$

15 – Gradients and differentiation

2 $-4x$ 3 -6 4 A 1



C $\frac{2}{(x+1)^2}; \frac{1}{2}$

5 A $f'(x) = \frac{1}{2}x^{-\frac{1}{2}}; f(9) = 3; f'(9) = \frac{1}{6}; f(9.2) = 3\frac{1}{30}$ B 3.03; 3.033

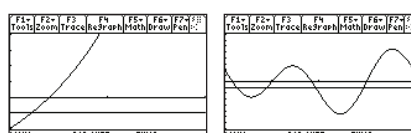
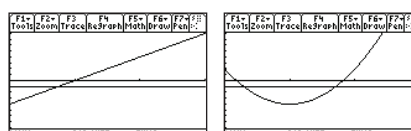
6 A $(x+h)^5$ B $(x+0.1)^5$ C $(x-0.1)^5$

7 A 283.5 B 202.5

8 $\frac{3(2x^2+2x-1)}{(x^2+3x+2)^2}; x = \frac{-\sqrt{3}-1}{2}$ or $x = \frac{\sqrt{3}-1}{2}$

9 A $k = \log_e 2$ B $(\log_e 2)e^{(\log_e 2)x} = (\log_e 2)2^x$

10 A 1 B 2 C 1 D 4 (one is a turning point)



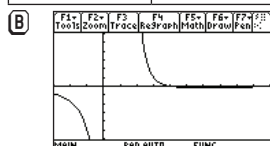


16 – Applications of differentiation to tangents to curves

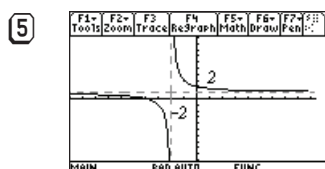
- ① 23 ② 0 ③ A $\frac{2}{e}$ B $y = \frac{2}{e}x$ ④ $y = -4x$ ⑤ 4; $\tan^{-1}4 = 75.96^\circ$
 ⑥ A 1; -1 B 45° ; 135° or -45° C $y = x - \frac{\pi}{6} + \frac{\sqrt{3}}{2}$; $y = -x + \frac{\pi}{6} + \frac{\sqrt{3}}{2}$
 ⑦ $(\sqrt{\log_e 2}, 0)$; $(-\sqrt{\log_e 2}, 0)$ ⑧ A 90° B 12.53°
 ⑨ A Solve $x^2 - 4 = 4 - x^2$ or substitute the given points in both equations B 4 C 75.96° D -4
 E -75.96° (or 104.04°) F 151.94° (or 28.08°)

17 – Graphs using calculus

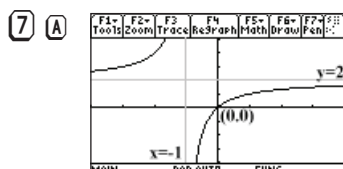
- ① (1, -4) minimum; (-1, 4) maximum
 ② A $f''(x)$ B $f'(x)$ C $f(x)$
 ③ A For reference $f(x) = \frac{4(x^2-4)}{(x-1)^2}$ and $f'(x) = \frac{-8(x-4)}{(x-1)^3}$
- | $f(x)$ | $f'(x)$ +ve, -ve or zero | Values of x |
|------------|----------------------------|----------------------------------|
| increasing | + | (0, 4) |
| decreasing | - | $(-\infty, 1)$ and $(4, \infty)$ |
| zero | +ve $x = 2$, -ve $x = -2$ | 2, 2 |
| maximum | zero | 4 |
| minimum | - | none |
| maximum | zero | 4 |



- ④ A $x^3 - 3x^2 - 24x + 80$ B (4, 0); (-2, 108)
 C maximum at (-2, 108); minimum at (4, 0) D $x < 1$ ($6x - 6 < 0$)

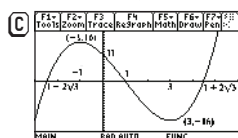


- ⑥ 0; 2



- B 0; all x excluding $x = -1$

- B A $x = 1, 1 \pm 2\sqrt{3}$ B $f'(x) = 3x^2 - 6x - 9$; maximum (-1, 16); minimum (3, -16)



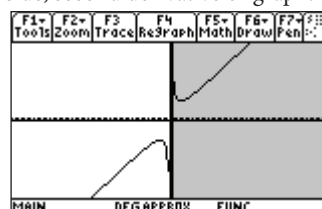
18 – Maximum and minimum tricks and traps

- ① A 2 at $x = 3$ B 4 at $x = 1$ C -1 at $x = 0$ D 5 at $x = 4$
 ② minima at (-2, -9) and (2, -9) and maximum at (0, 7)
 ③ minimum

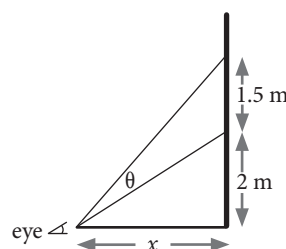
- ④ $-\left(\frac{1}{e}\right)$ ⑤ $a = -3$; $b = 2$
 ⑥ A $f'(x) = \frac{1}{2}x^2 - 4x + 1$ B 4
 ⑦ $x = 2$
 ⑧ $p = \frac{1}{2}$; $0 < p < 1$ so 0 and 1 are not solutions; p value is maximum because of the way the gradient on either side changes or using second derivative argument
 ⑨ A $x = 2$; $f(2) = 1$ B $A(0, \frac{1}{2})$; $B(2, 1)$; $C(7, 4\frac{1}{8})$
 C $x = 7$; $f(7) = 4\frac{1}{8}$
 ⑩ A $x = 600$ B $P = \$188.75$ C 140 to 1310 approximately

19 – Maximum and minimum 'wordies'

- ① A $10 = \frac{1}{2}r^2\theta$ B $P = 2r + r\theta$ C $P = 2r + \frac{20}{r}$ D $\sqrt{10}$; $4\sqrt{10}$
 (unshaded negative values are not feasible) E Gradient on either side, second derivative or graph.



- ② A B 15.83°



- ③ A amount of information presented must be positive or zero
 B $\frac{dy}{dx} = e^{-0.05x} (1 - \frac{x}{20})$ C 20 units D $y = \frac{20}{e}$; proportion 0.368
 ④ A 55 m B 5 m C 1 min D 0.3155 min or 18.93 secs
 E $0.6845 - 0.3155 = 0.369$ min

20 – Formulae in maximum and minimum 'Wordies'

- ② A x cm B $x(12-2x)(24-2x)$ C $0 < x < 6$
 D $192\sqrt{3}$ cm³ ($x = 6 - 2\sqrt{3}$)
 ③ A $P = 2r + 2s + \pi r$ B $A = 2rs + \frac{\pi r^2}{2}$ C $s = \frac{A - \frac{\pi r^2}{2}}{2r}$;
 $P = \frac{A}{r} + (2 + \frac{\pi}{2})r$ E $\sqrt{2A(4 + \pi)}$
 ④ A 10 cm³; $h = \frac{10}{\pi r^2}$ B $A = 2\pi r^2 \times \frac{10}{\pi r^2} + 2\pi r^2 = \frac{20}{r} + 2\pi r^2$
 C 1.17 cm D $h = 2.34$ cm
 ⑤ A $r = \sqrt{100 - \frac{h^2}{4}}$ C $V = 2418.4$ cm³; ($h = \frac{20\sqrt{3}}{3}$)

21 – Checklist for success

The columns should be ticked.

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